

ISO: Isospectral Optimization for RLVR

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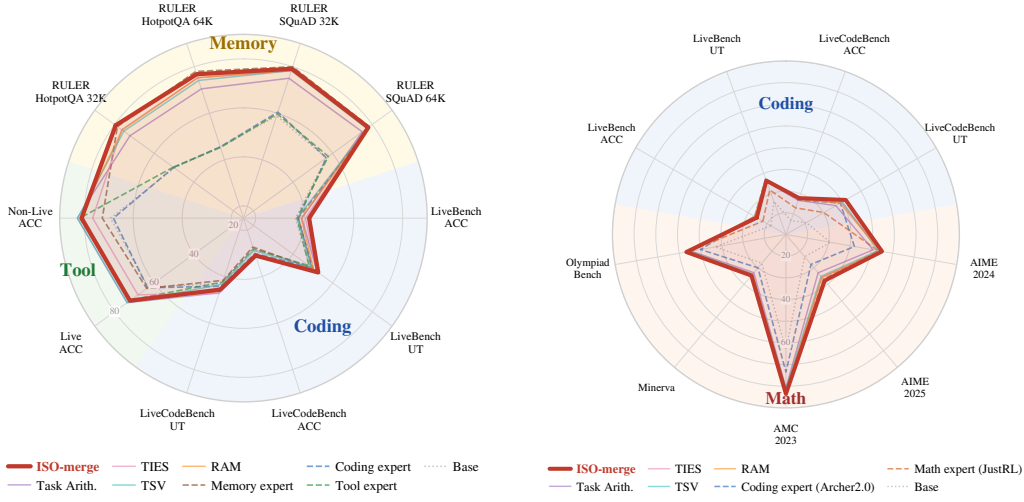


Figure 1: **ISO-merge enables training-free RL model merging.** (Left) Merging three domain experts (Coding, Tool, and Memory) based on Qwen2.5-7B-Instruct. (Right) Merging two domain experts (Coding and Math) based on DeepSeek-R1-Distill-Qwen-1.5B. ISO-merge consistently achieves the largest coverage area, demonstrating superior retention of individual expert capabilities compared to baseline methods.

Abstract

1 Reinforcement learning with verifiable rewards (RLVR) has become a key post-
 2 training phase for improving the reasoning ability of large language models and
 3 a central target of recent scaling efforts. While existing work has scaled RL data,
 4 environments, systems, and algorithms, the underlying optimization methods still
 5 largely follow practices inherited from pre-training and supervised fine-tuning.
 6 This default is increasingly questionable, as recent analyses suggest that RLVR
 7 follows a distinct optimization regime.

8 In this work, we study this regime through the SVD geometry of model weights. For
 9 a weight matrix $W = U\Sigma V^T$, we call a trajectory near-isospectral when its singular
 10 values remain close to the base spectrum over the observed training horizon. In the
 11 reasoning-oriented RLVR settings we analyze, key weight matrices exhibit small
 12 singular-value drift relative to SFT. Moreover, replacing the RL-trained spectrum
 13 with the base spectrum causes little performance degradation, while training only
 14 the spectrum fails to recover RLVR gains. These findings suggest that useful RLVR
 15 progress is carried mainly by singular-frame motion rather than spectral rescaling.

We further diagnose this motion on fixed top- r singular subspaces. On the truncated checkpoint transitions we analyze, target-subspace transport explains the observed movement substantially better than source-subspace fitting and one-sided alternatives. Motivated by this near-isospectral transport structure, we propose ISO, an isospectral optimization framework for RLVR. ISO contains two complementary methods: ISO-Merger, a data-free method that composes RLVR experts from a shared base by merging their Stiefel-tangent directions while preserving the base spectrum; and ISO-Optimizer, an online RLVR method that fixes each weight spectrum and updates only its Stiefel factors through manifold retraction. Experiments show that ISO-Merger improves data-free expert merging over standard weight-space baselines, while ISO-Optimizer improves RLVR stability and time-to-target compared with AdamW and Muon.

1 Introduction

Reinforcement learning (RL), and in particular reinforcement learning with verifiable rewards (RLVR), has become a key post-training stage for improving the reasoning ability of large language models. Recent efforts to push RL toward pretraining-comparable scale further suggest that RLVR is becoming an increasingly important phase for eliciting and scaling reasoning behavior [26]. Along this direction, recent progress has scaled RLVR across multiple axes, including larger verifiable datasets, richer reward environments, prolonged RL training, and more efficient RL systems [8, 13, 34]. Yet among these scaling axes, one component remains comparatively underdeveloped: the optimization method used to convert expensive rollouts and sparse rewards into parameter progress.

Most RLVR pipelines still inherit optimization practices from pre-training and supervised fine-tuning (SFT), for example by using AdamW as the default optimizer. This inheritance is increasingly questionable because RLVR differs from SFT in both its data-generation process and its learning signal: the policy samples its own trajectories [3], while rewards can be sparse, even effectively one-bit [18]. Recent studies further suggest that these differences lead to a distinct optimization trajectory. At the policy level, on-policy RL has been linked to improved generalization and reduced catastrophic forgetting [19]. At the parameter level, RLVR updates have been found to be sparse and off-principal in the original Euclidean parameter space, with limited spectral drift [14, 33]. This imbalance is striking: while optimization for pre-training has been extensively studied and repeatedly redesigned [32, 31, 12, 25], *RLVR—despite its high rollout cost and distinct optimization behavior—still largely relies on optimization recipes imported from pre-training and SFT.*

In this paper, we continue the study of RLVR’s distinct optimization regime by moving from Euclidean update statistics to the SVD geometry of model weights. Prior work asks how the raw update ΔW_t is oriented in parameter space; we instead decompose each weight matrix as

$$W_t = U_t \Sigma_t V_t^\top$$

and ask which component of this decomposition changes during RLVR. This separates spectral rescaling, encoded by the singular values Σ_t , from singular-frame motion, encoded by the orthonormal factors (U_t, V_t) .

This view reveals a simple empirical regularity. In the reasoning-oriented RLVR settings we study, key weight trajectories are *near-isospectral*: their singular values remain close to the base spectrum over the observed training horizon. This contrasts with SFT, which induces systematic spectral reshaping. Moreover, replacing the RL-trained singular values with the base singular values causes NO performance degradation, while training only the singular values fails to recover the RLVR gains. These interventions suggest that useful RLVR progress is carried mainly by singular-frame motion rather than spectral rescaling.

We then ask what kind of singular-frame motion best explains the observed checkpoints. On admissible truncated subspaces, our checkpoint diagnostics show that the observed transitions are better explained by target-subspace transport than by source-subspace fitting. In other words, the retained singular subspaces move through the ambient input and output spaces while the spectrum changes little. Our theory is used as a diagnostic organizer for this empirical picture: it gives finite-horizon spectral-drift bounds, gap-based subspace stability, and checkpoint-level comparisons among source-subspace, one-sided, and target-subspace transport models.

Motivated by this near-isospectral transport structure, we propose ISO, an isospectral optimization framework for RLVR. ISO has two complementary instantiations. **ISO-Merger** is a data-free method for composing multiple RLVR experts trained from the same base model. It maps each expert’s singular-frame displacement into the Stiefel tangent space of the shared base, merges the tangent directions, and reconstructs a model while preserving the base spectrum. **ISO-Optimizer** is an online RLVR method that fixes each weight spectrum and updates only its Stiefel factors through manifold retraction. In our experiments, ISO-Merger improves data-free expert merging over standard weight-space baselines, while ISO-Optimizer improves RLVR stability and time-to-target compared with AdamW and Muon.

Our contributions are summarized as follows:

- We provide an SVD-based empirical characterization of RLVR dynamics, showing that key weight matrices exhibit small finite-horizon singular-value drift while their dominant singular subspaces move nontrivially.
- We introduce checkpoint-level diagnostics for near-isospectral transport, showing that, on admissible truncated subspaces, observed RLVR transitions are better explained by target-subspace transport than by source-subspace fitting.
- We propose ISO-Merger, a data-free Stiefel-tangent method for composing RLVR experts from a shared base while preserving the base spectrum.
- We propose ISO-Optimizer, an online fixed-spectrum Stiefel optimizer for RLVR, improving stability and time-to-target in the studied RLVR settings.

2 Motivation: RLVR Calls for Better Learning Efficiency

[hz: to add]

3 Observation: Near-Isospectral Dynamics in RLVR

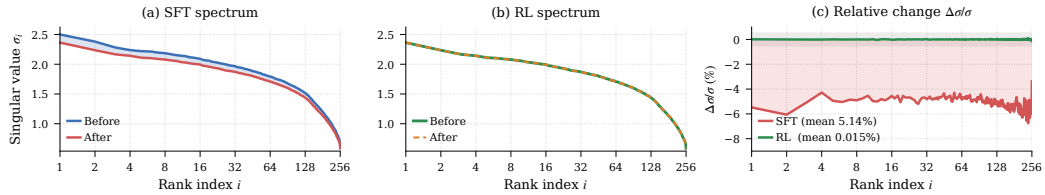


Figure 2: **Spectral dynamics under SFT vs. RL post-training.** **Top:** Singular value profiles before and after training. SFT produces a consistent spectral shrinkage, while RL preserves the singular spectrum. **Bottom:** Relative change in singular values. Under RL, the relative drift stays close to zero across most ranks, whereas SFT induces substantially larger deviations.

We begin with an empirical observation about the SVD geometry of RLVR-trained weights. For a weight matrix $W_t \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, let

$$W_t = U_t \Sigma_t V_t^\top$$

denote its singular value decomposition, and let $\sigma(W_t)$ be the full singular-value vector. We call a trajectory *near-isospectral* if its singular values remain close to those of the pre-RL checkpoint over the observed training horizon. This section is purely empirical: we measure spectral drift, intervene on the spectrum, and test whether training the spectrum alone is sufficient for RLVR.

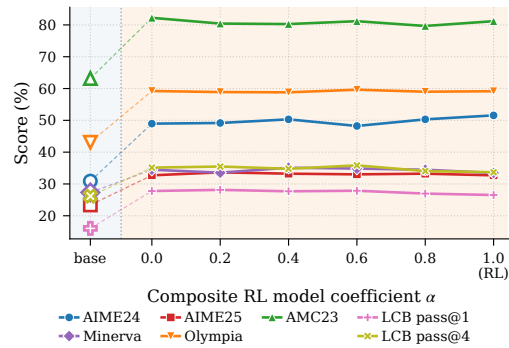


Figure 3: **Invariant task performance under singular value replacement.** The task accuracy remains stable across different interpolation coefficients α .

In this section, we use DS-1.5B as the case study, as one long-horizon run exist in [13] with 3k RL steps. We compare the base (Qwen/Qwen2.5-Math-1.5B), SFT version (deepseek-ai/DeepSeek-R1-Distill-Qwen-1.5B) and RL version (nvidia/Nemotron-Research-Reasoning-Qwen-1.5B).

3.1 Spectral drift is small under RLVR

We first compare the spectral evolution of RLVR and SFT. For each matrix, we measure normalized singular-value drift relative to the pre-training or pre-RL checkpoint:

$$\Delta_\sigma(t) = \frac{\|\sigma(W_t) - \sigma(W_0)\|_2}{\|\sigma(W_0)\|_2}.$$

As shown in Figure 2, RLVR produces substantially smaller spectral drift than SFT. In the studied reasoning-oriented RLVR runs, the singular-value profiles before and after RL nearly overlap, whereas SFT induces systematic spectral reshaping. This suggests that RLVR does not primarily improve the model by rescaling the singular spectrum.

Connection to concurrent work. This observation of spectral stability is consistent with recent findings in the literature. *The Path Not Taken* [33] also reports similar "off-principal" with stable spectral dynamics, reinforcing the hypothesis that RL updates are geometrically distinct from the spectral reshaping characteristic of pre-training.

3.2 Rebased the spectrum preserves RLVR performance

Small spectral drift alone does not imply that the spectrum is functionally irrelevant. We therefore perform a direct intervention. Let

$$W_{\text{RL}} = U_{\text{RL}} \Sigma_{\text{RL}} V_{\text{RL}}^\top$$

be an RL-trained checkpoint and let Σ_{base} be the spectrum of the corresponding pre-RL checkpoint. We construct

$$\widetilde{W}(\alpha) = U_{\text{RL}}(\alpha \Sigma_{\text{RL}} + (1 - \alpha) \Sigma_{\text{base}}) V_{\text{RL}}^\top, \quad \alpha \in [0, 1].$$

Here $\alpha = 1$ recovers the RL checkpoint, while $\alpha = 0$ fully replaces the RL-trained spectrum with the base spectrum while keeping the RL-trained singular frames fixed.

Figure 3 show that this replacement causes little performance degradation on both math and code evaluations. In particular, setting $\alpha = 0$ retains the RL performance gains, indicating that the behavioral improvement is not primarily stored in the learned singular values. Instead, the useful change is largely carried by the learned singular frames $(U_{\text{RL}}, V_{\text{RL}})$.

3.3 Training only the spectrum does not recover RLVR gains

As a complementary test, we ask whether the singular values alone are sufficient degrees of freedom for RLVR. We freeze the singular frames (U, V) of each weight matrix and train only the diagonal entries of Σ . This isolates spectral rescaling while removing singular-frame motion.

As shown in Table ??, Σ -only training fails to recover the gains of RLVR, whereas ISO, which fixes Σ and updates the singular frames, improves downstream performance. Together with the replacement experiment, this provides a two-sided intervention: changing the spectrum is not necessary to preserve RLVR performance, and optimizing only the spectrum is not sufficient to obtain RLVR performance.

4 Checkpoint-Level Diagnostics of Singular-Frame Transport

Section ?? shows that RLVR changes the singular spectrum little and that replacing the RL-trained spectrum with the base spectrum causes little performance degradation. We now ask a complementary question: if useful RLVR progress is not primarily stored in the singular values, what kind of singular-frame motion best explains the observed checkpoints?

Our goal in this section is diagnostic rather than predictive. We do not claim to derive RLVR dynamics from the RL objective. Instead, we compare several geometric reconstruction classes on observed checkpoint transitions and ask which class provides the best fit.

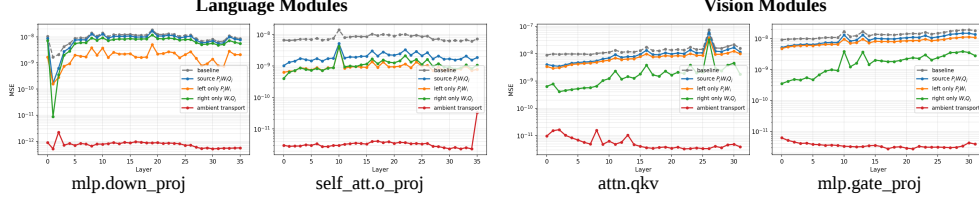


Figure 4: **Geometric decomposition of RL updates** ($W_0 \rightarrow W_2$ at truncation rank $r = 0.9q$). The ambient-transport class fits each post-RL checkpoint $\sim 10^2 \times$ more accurately than the source-subspace class or either one-sided hybrid, showing that **RL updates are dominated by ambient transport rather than in-subspace mixing**. This dominance is insensitive to the truncation rank; full comparisons across all layers, transitions, and $r \in [0.2q, 0.9q]$ are deferred to the Appendix.

146 **Top- r diagnostic subspaces.** For a weight matrix $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, let $q = \min(d_{\text{out}}, d_{\text{in}})$. For
 147 singular-value drift and spectrum-replacement experiments, we use the full singular-value vector. For
 148 subspace diagnostics, however, we fix a truncation rank $r < q$ and compute the top- r SVD

$$W_t^{(r)} = U_t \Sigma_t V_t^\top, \quad U_t \in \text{St}(d_{\text{out}}, r), \quad V_t \in \text{St}(d_{\text{in}}, r).$$

149 All source-vs-target transport diagnostics below are computed on this fixed top- r truncation. This
 150 truncation is essential: if a full-rank square SVD were used, then $U_t U_t^\top = I$ and $V_t V_t^\top = I$, so
 151 source-subspace projection would become vacuous. The diagnostic therefore focuses on whether the
 152 dominant singular subspaces move through the ambient input/output spaces.

153 Unless otherwise stated, we use $r = 0.9q$ in the main text. Appendix ?? reports rank sensitivity over
 154 $r \in \{0.2q, 0.4q, 0.6q, 0.8q, 0.9q\}$, together with retained energy and boundary-gap diagnostics.

155 **Checkpoint reconstruction classes.** Consider a transition from checkpoint j to checkpoint i , with
 156 top- r decompositions

$$W_j^{(r)} = U_j \Sigma_j V_j^\top, \quad W_i^{(r)} = U_i \Sigma_i V_i^\top.$$

157 We compare four reconstruction classes.

158 First, the *source-subspace fit* asks whether the target checkpoint can be represented inside the source
 159 checkpoint’s retained left and right singular subspaces:

$$\widehat{W}_{\text{src}} = U_j \left(U_j^\top W_i^{(r)} V_j \right) V_j^\top.$$

160 Equivalently, this is $P_j W_i^{(r)} Q_j$, where $P_j = U_j U_j^\top$ and $Q_j = V_j V_j^\top$.

161 Second, the one-sided hybrid fits test whether only one side of the singular subspace has moved:

$$\widehat{W}_L = U_j U_j^\top W_i^{(r)}, \quad \widehat{W}_R = W_i^{(r)} V_j V_j^\top.$$

162 The left hybrid keeps the source left subspace but allows the target right orientation; the right hybrid
 163 is symmetric.

164 Third, the *target-subspace transport* fit keeps the target singular subspaces but substitutes the source
 165 spectrum:

$$\widehat{W}_{\text{tgt}} = U_i \Sigma_j V_i^\top.$$

166 This model captures a near-isospectral transition in which the dominant singular frames move while
 167 the retained spectral energy is preserved.

168 For each reconstruction $\widehat{W}$, we report the normalized Frobenius error

$$e(\widehat{W}) = \frac{\|W_i^{(r)} - \widehat{W}\|_F}{\|W_i^{(r)}\|_F}.$$

169 Thus,

$$e_{\text{src}} = \frac{\|W_i^{(r)} - \widehat{W}_{\text{src}}\|_F}{\|W_i^{(r)}\|_F},$$

$$e_L = \frac{\|W_i^{(r)} - \widehat{W}_L\|_F}{\|W_i^{(r)}\|_F}, \quad e_R = \frac{\|W_i^{(r)} - \widehat{W}_R\|_F}{\|W_i^{(r)}\|_F},$$

and

$$e_{\text{tgt}} = \frac{\|W_i^{(r)} - \widehat{W}_{\text{tgt}}\|_F}{\|W_i^{(r)}\|_F} = \frac{\|\Sigma_i - \Sigma_j\|_F}{\|\Sigma_i\|_F}.$$

The last equality follows immediately from

$$W_i^{(r)} - \widehat{W}_{\text{tgt}} = U_i(\Sigma_i - \Sigma_j)V_i^\top.$$

Therefore, target-subspace transport has small error exactly when the retained spectrum changes little after expressing the checkpoint in its target singular frames.

Two-stage RL diagnostic. We evaluate these diagnostics on a two-stage RL pipeline starting from a shared base checkpoint W_0 . Stage I performs RL on one verifiable task to obtain W_1 , and Stage II continues RL on a different task to obtain W_2 . This gives three checkpoint transitions:

$$W_0 \rightarrow W_1, \quad W_1 \rightarrow W_2, \quad W_0 \rightarrow W_2.$$

For each transition and each 2D weight matrix in the language and vision modules, we compute the four normalized errors

$$e_{\text{src}}, \quad e_L, \quad e_R, \quad e_{\text{tgt}}.$$

We also report the raw drift

$$e_{\text{drift}} = \frac{\|W_i^{(r)} - W_j^{(r)}\|_F}{\|W_i^{(r)}\|_F}$$

as a scale reference, but the model-class comparison is determined by the relative ordering of the reconstruction errors.

Results. Figure 4 shows the reconstruction errors across layers for both language modules and vision modules. The source-subspace fit remains substantially worse than the target-subspace transport fit, indicating that the target checkpoint cannot be well explained by staying inside the source checkpoint’s retained singular subspaces. The one-sided hybrids provide a finer diagnostic: when both e_L and e_R remain much larger than e_{tgt} , the motion is two-sided; when only one hybrid is large, the transition is mostly left- or right-subspace movement.

Across the analyzed transitions, e_{tgt} is consistently the smallest or near-smallest error. Since e_{tgt} reduces to retained spectral mismatch, this result supports the near-isospectral transport picture: the dominant singular subspaces move through the ambient input/output spaces, while the retained spectrum changes little.

Takeaway. The checkpoint diagnostics do not prove that RLVR must follow target-subspace transport. They show that, on the analyzed fixed top- r transitions, target-subspace transport provides a substantially better checkpoint-level explanation than source-subspace fitting and one-sided alternatives. This empirical geometry motivates the fixed-spectrum methods introduced next.

5 Isospectral Optimization Framework for RLVR

5.1 A Shared Fixed-Spectrum Tangent Parameterization

For a matrix $W = U\Sigma V^\top$, with $U \in \text{St}(m, q)$, $V \in \text{St}(n, q)$, and fixed $\Sigma \in \mathbb{R}^{q \times q}$, the first-order weight motion induced by Stiefel-factor displacements (ξ_U, ξ_V) is

$$\mathcal{T}_{U,V,\Sigma}(\xi_U, \xi_V) := \xi_U \Sigma V^\top + U \Sigma \xi_V^\top,$$

where

$$\xi_U \in T_U \text{St}(m, q), \quad \xi_V \in T_V \text{St}(n, q).$$

Equivalently,

$$U^\top \xi_U + \xi_U^\top U = 0, \quad V^\top \xi_V + \xi_V^\top V = 0.$$

203 This tangent map has zero first-order singular-value variation:

$$\text{diag}(U^\top \mathcal{T}_{U,V,\Sigma}(\xi_U, \xi_V)V) = 0.$$

204 Indeed,

$$U^\top \mathcal{T}_{U,V,\Sigma}(\xi_U, \xi_V)V = (U^\top \xi_U)\Sigma + \Sigma(\xi_V^\top V),$$

205 and both terms have zero diagonal because $U^\top \xi_U$ and $V^\top \xi_V$ are skew-symmetric.

206 ISO uses this same fixed-spectrum tangent geometry in two settings: ISO-Merger composes offline
207 expert displacements in this tangent space, while ISO-Optimizer performs online RLVR by optimizing
208 U, V and retracting them back to the Stiefel manifold.

209 5.2 Pareto-Aware Stiefel-Tangent Merging (PAST-Merge)

210 **Setup.** Let $W_{\text{pre}} \in \mathbb{R}^{m \times n}$ be a pretrained 2D weight matrix, and let $\{W_i\}_{i=1}^T$ be the corresponding
211 matrices from T RL fine-tuned experts. Motivated by the rotation-only structure observed in the
212 preceding analysis, we represent each expert as preserving the base spectrum while changing the left
213 and right singular frames:

$$W_i \approx U_i \Sigma_{\text{pre}} V_i^\top, \quad U_i \in \text{St}(k, m), \quad V_i \in \text{St}(k, n), \quad \Sigma_{\text{pre}} \in \mathbb{R}^{k \times k}. \quad (1)$$

214 Our goal is to construct a merged matrix

$$W_\star = U_\star \Sigma_{\text{pre}} V_\star^\top \quad (2)$$

215 that preserves the first-order weight effect of each expert while remaining on the Stiefel product
216 manifold.

217 **Stiefel tangent representation.** For an anchor $U_{\text{pre}} \in \text{St}(k, m)$, the tangent space is

$$T_{U_{\text{pre}}} \text{St}(k, m) = \{\xi \in \mathbb{R}^{m \times k} : U_{\text{pre}}^\top \xi + \xi^\top U_{\text{pre}} = 0\}. \quad (3)$$

218 Given an ambient displacement $\delta_{u,i} = U_i - U_{\text{pre}}$, we use its Euclidean projection onto the tangent
219 space as a first-order Stiefel displacement:

$$\xi_{u,i} = \Pi_{U_{\text{pre}}}(\delta_{u,i}) \triangleq \delta_{u,i} - U_{\text{pre}} \text{sym}(U_{\text{pre}}^\top \delta_{u,i}), \quad \text{sym}(A) = \frac{1}{2}(A + A^\top). \quad (4)$$

220 The same construction gives $\xi_{v,i} \in T_{V_{\text{pre}}} \text{St}(k, n)$ for the right singular frame. To map a tangent
221 update back to the Stiefel manifold, we use the chordal retraction

$$\text{Retr}_{U_{\text{pre}}}(\xi) = \text{Polar}(U_{\text{pre}} + \xi), \quad (5)$$

222 where $\text{Polar}(X) = PQ^\top$ for the thin SVD $X = PSQ^\top$.

223 **Top- k singular-mode truncation.** Let $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_k$ be the diagonal of Σ_{pre} , ordered by
224 the base spectrum. We restrict the merge to the leading $k_\rho = \lceil \rho k \rceil$ singular modes (with $\rho \in (0, 1]$ a
225 hyperparameter, in our final model $\rho = 0.9$). Concretely, the columns of each tangent indexed by
226 $r \geq k_\rho$ are zeroed out:

$$\tilde{\xi}_{u,i}[:, r] = \begin{cases} \xi_{u,i}[:, r], & r < k_\rho, \\ \mathbf{0}, & r \geq k_\rho, \end{cases} \quad \tilde{\xi}_{v,i}[:, r] \text{ analogously}. \quad (6)$$

227 This restricts the merged update to the high-energy subspace of the base spectrum, where the
228 rotation-only signal is strongest. Tail modes carry weak signal but contribute disproportionately to the
229 cross-bilinear interference term, and discarding them sharply reduces interference without changing
230 the top- k retention property derived below.

231 **First-order weight effect and Gram matrix.** For each expert, the first-order effect of the truncated
232 tangent pair $(\tilde{\xi}_{u,i}, \tilde{\xi}_{v,i})$ on the weight matrix is

$$g_i = \tilde{\xi}_{u,i} \Sigma_{\text{pre}} V_{\text{pre}}^\top + U_{\text{pre}} \Sigma_{\text{pre}} \tilde{\xi}_{v,i}^\top. \quad (7)$$

233 We collect pairwise inner products of these directions in the Gram matrix

$$G_{ij} = \langle g_i, g_j \rangle_F, \quad G \in \mathbb{R}^{T \times T}. \quad (8)$$

234 This matrix measures how strongly the experts' first-order effects align or interfere on the top- k_ρ
235 subspace.

236 **Closed-form retention coefficients with clipping.** Let $c \in \mathbb{R}^T$ be the merging coefficients. The
 237 merged first-order direction is $g_\star = \sum_i c_i g_i$, and the retention of expert i along its own direction is

$$r_i = \frac{\langle g_\star, g_i \rangle_F}{\langle g_i, g_i \rangle_F} = \frac{(Gc)_i}{G_{ii}}. \quad (9)$$

238 Requiring unit retention for all experts gives the linear system

$$Gc^\star = \text{diag}(G), \quad c^\star = (G + \eta I)^{-1} \text{diag}(G), \quad (10)$$

239 where $\eta = 10^{-12}$ is a small ridge for numerical stability. When two experts are nearly collinear (G
 240 ill-conditioned), the closed-form solution can produce coefficients with extreme magnitudes or wrong
 241 signs that subtract one expert from another. We therefore clip $c^\star$ entrywise:

$$c_i^\star \leftarrow \text{clip}(c_i^\star, c_{\min}, c_{\max}), \quad [c_{\min}, c_{\max}] = [0, 1.5]. \quad (11)$$

242 The lower bound $c_{\min} = 0$ rules out expert negation; the upper bound $c_{\max} = 1.5$ caps over-
 243 amplification.

244 **Stiefel-tangent re-projection.** The truncation in Eq. (6) zeros columns $r \geq k_\rho$, which generally
 245 breaks the skew constraint $U_{\text{pre}}^\top \xi + \xi^\top U_{\text{pre}} = 0$. After summing the clipped coefficients, we re-apply
 246 the Stiefel-tangent projection to restore tangency:

$$\xi_{u,\star} \leftarrow \Pi_{U_{\text{pre}}} \left(\sum_{i=1}^T c_i^\star \tilde{\xi}_{u,i} \right), \quad \xi_{v,\star} \leftarrow \Pi_{V_{\text{pre}}} \left(\sum_{i=1}^T c_i^\star \tilde{\xi}_{v,i} \right). \quad (12)$$

247 The reconstructed matrix is

$$W_\star = \text{Polar}(U_{\text{pre}} + \xi_{u,\star}) \Sigma_{\text{pre}} \text{Polar}(V_{\text{pre}} + \xi_{v,\star})^\top. \quad (13)$$

248 **Embedding, LM head, and 1D parameters.** The Stiefel-tangent rule in Eqs. (4)–(13) is applied
 249 to every 2D parameter matrix, including the token embedding and LM head when they are treated as
 250 matrix parameters. For 1D parameters, such as normalization scales and biases, we use a numerically
 251 stable Welford rolling mean of task vectors:

$$\delta^{(0)} = 0, \quad \delta^{(i)} = \delta^{(i-1)} + \frac{1}{i} \left[(\theta_i - \theta_{\text{pre}}) - \delta^{(i-1)} \right], \quad i = 1, \dots, T. \quad (14)$$

252 The merged 1D parameter is then $\theta_\star = \theta_{\text{pre}} + \delta^{(T)}$.

253 5.3 Evaluation Protocol

254 5.4 Experimental Results

Table 1: **Main results on Qwen2.5-7B with 3 RL experts.** Coding: pass accuracy (ACC) and unit-test accuracy (UT) on LiveBench (LB) and LiveCodeBench (LCB). Tool Use: Live and Non-Live averages following the BFCL v4 evaluation protocol. Memory: RULER HotpotQA and SQuAD at 7K–64K context lengths under the Recurrent setting. Best merged result per column in **bold**.

Method	Coding				Tool Use		Memory					
	LB		LCB v2		Live	Non-Live	HotpotQA		SQuAD		Avg	
	ACC	UT	ACC	UT	ACC	ACC	32K	64K	32K	64K		
Base [17]	36.25 ± 0.86	48.27 ± 1.44	28.31 ± 0.51	43.08 ± 0.82	62.96	69.26	49.87 ± 1.45	45.54 ± 0.58	58.90 ± 1.07	56.77 ± 0.61	49.91	
RLVR-Coder [23]	37.42 ± 1.22	49.42 ± 1.60	30.17 ± 0.99	44.24 ± 0.99	63.37	68.07	50.21 ± 0.16	45.61 ± 0.33	60.51 ± 0.70	56.84 ± 0.42	50.56	
RLVR-Tool [16]	36.87 ± 1.79	48.08 ± 0.82	28.20 ± 1.95	42.43 ± 1.90	71.83	81.82	50.39 ± 0.68	45.33 ± 0.26	59.73 ± 1.18	57.86 ± 0.52	52.37	
RLVR-Memory [28]	37.32 ± 1.37	50.05 ± 1.61	27.41 ± 2.99	41.70 ± 4.63	63.82	72.61	78.60 ± 0.54	77.95 ± 0.29	79.98 ± 0.38	78.52 ± 0.22	60.79	
Task Arithmetic [9]	38.85 ± 0.70	52.18 ± 0.40	31.55 ± 0.18	47.06 ± 0.58	73.73	82.86	72.22 ± 0.22	70.49 ± 0.10	75.03 ± 0.29	74.95 ± 0.32	61.89	
TIES [27]	40.30 ± 2.18	52.04 ± 2.50	31.46 ± 0.35	47.33 ± 0.30	68.24	76.54	76.42 ± 0.36	75.24 ± 0.14	78.27 ± 0.46	77.86 ± 0.59	62.37	
TSV [5]	36.96 ± 0.86	49.23 ± 0.59	28.92 ± 1.79	43.37 ± 3.25	73.64	82.69	75.37 ± 0.29	74.10 ± 0.30	78.30 ± 0.17	77.26 ± 0.78	61.98	
RAM [11]	38.59 ± 0.79	50.38 ± 1.53	31.31 ± 0.28	46.33 ± 0.30	72.56	81.89	76.17 ± 0.20	75.29 ± 0.49	78.42 ± 0.47	77.83 ± 0.46	62.88	
Ours	41.81 ± 2.00	52.73 ± 1.57	31.06 ± 1.88	45.69 ± 3.06	72.32	81.07	79.46 ± 0.78	76.77 ± 0.27	79.10 ± 0.14	78.01 ± 0.32	63.80	

Algorithm 1 Pareto-Aware Stiefel-Tangent Merging (PAST-Merge)

Require: Pretrained model θ_{pre} ; expert models $\{\theta_i\}_{i=1}^T$; top- k keep ratio $\rho \in (0, 1]$; clip range $[c_{\min}, c_{\max}]$; ridge $\eta \geq 0$

Ensure: Merged model θ_{PAST}

- 1: Initialize $\theta_{\text{PAST}} \leftarrow \theta_{\text{pre}}$
- // Stage A: Stiefel-tangent merge for 2D parameters**
- 2: **for** each 2D matrix $W_{\text{pre}}^{(\ell)} \in \mathbb{R}^{m_\ell \times n_\ell}$ **do**
- 3: $(U_{\text{pre}}, \Sigma_{\text{pre}}, V_{\text{pre}}) \leftarrow \text{ThinSVD}(W_{\text{pre}}^{(\ell)})$
- 4: $k_\rho \leftarrow \lceil \rho \cdot k \rceil$ $\triangleright$ top- k rank cutoff
- 5: **for** $i = 1, \dots, T$ **do**
- 6: $(U_i, \hat{\Sigma}_i, V_i) \leftarrow \text{ThinSVD}(W_i^{(\ell)})$
- 7: $(U_i, V_i) \leftarrow \text{SignAlign}(U_{\text{pre}}, U_i, V_i)$ $\triangleright$ fix SVD sign ambiguity
- 8: $\xi_{u,i} \leftarrow \Pi_{U_{\text{pre}}}(U_i - U_{\text{pre}}), \quad \xi_{v,i} \leftarrow \Pi_{V_{\text{pre}}}(V_i - V_{\text{pre}})$
- 9: $\tilde{\xi}_{u,i} \leftarrow \xi_{u,i}, \quad \tilde{\xi}_{u,i}[:, k_\rho:] \leftarrow \mathbf{0}$ $\triangleright$ top- k truncation
- 10: $\tilde{\xi}_{v,i} \leftarrow \xi_{v,i}, \quad \tilde{\xi}_{v,i}[:, k_\rho:] \leftarrow \mathbf{0}$
- 11: $g_i \leftarrow \tilde{\xi}_{u,i} \Sigma_{\text{pre}} V_{\text{pre}}^\top + U_{\text{pre}} \Sigma_{\text{pre}} \tilde{\xi}_{v,i}^\top$
- 12: **end for**
- 13: $G_{ij} \leftarrow \langle g_i, g_j \rangle_F$ for all $i, j \in \{1, \dots, T\}$
- 14: $c^* \leftarrow (G + \eta I)^{-1} \text{diag}(G)$ $\triangleright$ closed-form unit-retention solution
- 15: $c_i^* \leftarrow \text{clip}(c_i^*, c_{\min}, c_{\max})$ for $i = 1, \dots, T$ $\triangleright$ prevent negation & over-amplification
- 16: $\xi_{u,*} \leftarrow \Pi_{U_{\text{pre}}}(\sum_{i=1}^T c_i^* \tilde{\xi}_{u,i})$ $\triangleright$ re-project after truncation
- 17: $\xi_{v,*} \leftarrow \Pi_{V_{\text{pre}}}(\sum_{i=1}^T c_i^* \tilde{\xi}_{v,i})$
- 18: $U_* \leftarrow \text{Polar}(U_{\text{pre}} + \xi_{u,*}), \quad V_* \leftarrow \text{Polar}(V_{\text{pre}} + \xi_{v,*})$
- 19: $W_{\text{PAST}}^{(\ell)} \leftarrow U_* \Sigma_{\text{pre}} V_*^\top$
- 20: **end for**
- // Stage B: Welford mean for 1D parameters**
- 21: **for** each 1D parameter $v_{\text{pre}} \in \mathbb{R}^d$ **do**
- 22: $\delta \leftarrow \mathbf{0}$
- 23: **for** $i = 1, \dots, T$ **do**
- 24: $\delta \leftarrow \delta + \frac{1}{i} [(v_i - v_{\text{pre}}) - \delta]$
- 25: **end for**
- 26: $v_{\text{PAST}} \leftarrow v_{\text{pre}} + \delta$
- 27: **end for**
- 28: **return** θ_{PAST}

255 6 The Isospectral Stiefel Optimizer

256 **Local fixed-spectrum rationale.** For $W = U\Sigma V^\top$, a first-order perturbation changes singular
 257 values according to the diagonal of $U^\top \Delta W V$. If Σ is fixed and we move only through Stiefel tangent
 258 directions,

$$\Delta W = \xi_U \Sigma V^\top + U \Sigma \xi_V^\top,$$

259 where $U^\top \xi_U$ and $V^\top \xi_V$ are skew-symmetric, then

$$\text{diag}(U^\top \Delta W V) = 0.$$

260 Thus the fixed-spectrum parameterization removes explicit first-order spectral-drift coordinates. This
 261 does not imply that the restriction is always beneficial; it provides a local geometric rationale for ISO
 262 under the empirical near-isospectral regime identified above.

263 Motivated by our observation that RL post-training is approximately isospectral and is dominated
 264 by ambient transport of feature subspaces, we propose an optimizer that explicitly enforces this
 265 geometry by keeping the singular spectrum of each weight matrix fixed throughout training. We
 266 call this method the **Isospectral Stiefel Optimizer (ISO)**. ISO parameterizes each weight matrix
 267 by its Stiefel factors (U, V) and performs updates only on these factors, while holding the spectrum
 268 Σ constant for the entire post-training run. Here, we introduce three key components of ISO. The
 269 complete procedure is summarized in Algorithm ??.

Table 2: **Main results on DeepSeek-R1-Distill-Qwen-1.5B with 2 RL experts.** Coding: pass accuracy (ACC) and unit-test accuracy (UT) on LiveBench (LB) and LiveCodeBench (LCB). Math: ACC on AIME 2024, AIME 2025, AMC 2023, Minerva, and OlympiadBench. Best merged result per column in **bold**.

Method	Coding					Math						
	LB		LCB v5		Avg	AIME 2024	AIME 2025	AMC 2023	Minerva	OlympiadBench	Avg	Total Avg
	ACC	UT	ACC	UT		Avg@32	Avg@32	Avg@8	Avg@4	Avg@4		
Base [4]	17.99 ± 0.29	26.24 ± 0.53	16.82 ± 0.28	20.85 ± 0.18	20.47	30.90 ± 1.15	23.40 ± 0.86	63.15 ± 0.38	27.33 ± 0.43	43.11 ± 0.26	37.58	29.03
Archer2.0 [22]	26.66 ± 0.45	37.26 ± 0.55	26.90 ± 0.21	38.99 ± 0.37	32.45	42.05 ± 0.43	28.02 ± 0.44	73.44 ± 0.50	30.09 ± 0.23	49.99 ± 0.65	44.72	38.59
JustRL [6]	22.23 ± 0.23	31.85 ± 0.74	23.05 ± 0.21	30.18 ± 0.31	26.83	53.16 ± 0.44	36.84 ± 0.73	82.78 ± 0.51	34.71 ± 0.50	55.67 ± 0.42	52.63	39.73
Task Arithmetic [9]	26.22 ± 0.22	35.82 ± 0.11	26.70 ± 0.21	36.58 ± 0.32	31.33	50.83 ± 0.39	33.09 ± 1.16	80.82 ± 0.51	33.00 ± 0.34	54.31 ± 0.15	50.41	40.87
TIES [27]	26.37 ± 0.73	36.75 ± 0.90	27.62 ± 0.13	39.45 ± 0.15	32.55	54.51 ± 0.72	35.76 ± 0.30	82.13 ± 0.14	33.76 ± 0.48	55.36 ± 0.15	52.31	42.43
TSV [5]	26.40 ± 0.40	36.83 ± 0.15	27.61 ± 0.17	40.01 ± 0.26	32.71	53.16 ± 0.57	35.17 ± 0.72	80.97 ± 0.92	33.88 ± 0.46	54.68 ± 0.27	51.57	42.14
RAM [11]	26.71 ± 0.29	36.99 ± 0.47	27.18 ± 0.63	38.83 ± 0.67	32.43	54.20 ± 0.55	35.80 ± 0.71	82.28 ± 0.38	34.10 ± 0.34	55.54 ± 0.25	52.38	42.41
Ours	25.52 ± 0.08	36.42 ± 0.55	27.84 ± 0.13	41.84 ± 0.31	32.91	55.00 ± 0.52	37.81 ± 0.31	83.53 ± 1.02	34.77 ± 0.23	56.64 ± 0.05	53.55	43.23

(1) **Isospectral parameterization.** For each weight matrix $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, we compute an initial reduced SVD $W_0 = U_0 \Sigma_0 V_0^\top$ and freeze the spectrum by setting $\Sigma_{\text{fixed}} \leftarrow \Sigma_0$. We then parameterize the trainable weight as

$$W(U, V) = U \Sigma_{\text{fixed}} V^\top,$$

where $U \in \text{St}(d_{\text{out}}, r)$ and $V \in \text{St}(d_{\text{in}}, r)$. Since U and V have orthonormal columns, the singular values of $W(U, V)$ are identically Σ_{fixed} . This removes degrees of freedom that would otherwise be spent reshaping a spectrum that is empirically stable under RL post-training.

(2) **Factor gradients.** Let $G_W = \nabla_W \mathcal{L}(W)$ denote the Euclidean gradient with respect to the reconstructed weight. With Σ_{fixed} held constant, the gradients of the Stiefel factors follow from the chain rule:

$$G_U = G_W V \Sigma_{\text{fixed}}, \quad G_V = G_W^\top U \Sigma_{\text{fixed}}.$$

(3) **Retraction to the manifold.** After a Euclidean optimizer step, we retract the updated factors back onto the Stiefel manifold using the polar factor, implemented via the matrix sign operator $\text{msign}(\cdot)$:

$$\text{Retract}(X) \triangleq \text{msign}(X).$$

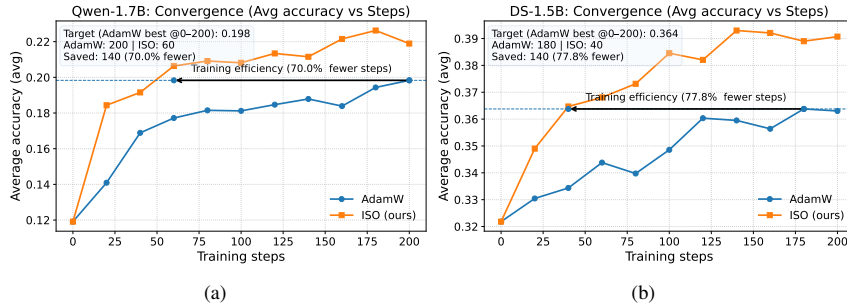


Figure 5: **Faster convergence with ISO.** Average evaluation accuracy versus training steps for (a) Qwen3-1.7B-Base and (b) DS-R1-Distill-Qwen-1.5B. ISO reaches the target accuracy in substantially fewer steps than AdamW.

7 Experiments

Experiments Setups. [zhizhou: These part will be need to update]

We consider two representative base models: **Qwen3-1.7B-Base** [21] and **DeepSeek-R1-Distill-Qwen-1.5B** [4]. The former is a widely used community model, while the latter is optimized for longer responses and better reflects practical, industry-style post-training. As our primary baseline optimizer, we use AdamW, a strong and widely used optimizer by the community.

Implementation details. We build our training pipeline on top of the VERL [20] framework. Following prior work [18], we use DEEPMATH [8] as the RL training set. Unless stated otherwise, we use a global batch size of 128 and an on-policy minibatch size of 128. For each prompt, we sample 12 rollouts. We set the learning rate to 10^{-6} for all methods to ensure a fair comparison. For ISO, we compute the SVD in fp64 for numerical stability; see Section D for a detailed discussion of precision choices.

Evaluation details. To measure mathematical reasoning performance, we evaluate on five benchmarks: AIME24 [29], AIME25 [30], AMC23, Minerva Math [10], and OlympiadBench [7]. We use evaluation framework in POLARIS [1] and use the evaluation setting used in *The Path Not Taken* [33], we use temperature 1.0, top- p 0.8, and a maximum generation length of 4096 for Qwen3-1.7B-Base and 9216 for DS-R1-Distill-Qwen-1.5B.

Table 3: Main results comparing ISO against AdamW, SGD, and Muon on RL post-training with DeepMath. We report accuracy on five math reasoning benchmarks. ISO results are shaded.

Model	Method	AIME24 avg@32	AIME25 avg@32	AMC23 avg@4	Minerva avg@4	Olympia avg@8	Avg.
Qwen3-1.7B-Base	Base	3.96	2.40	21.99	13.14	18.07	11.91
	AdamW	8.23	5.00	34.49	23.07	28.37	19.83
	Muon	-	-	-	-	-	-
	ISO	9.27	5.73	36.14	26.10	32.22	21.89
Qwen3-4B-Base	Base	-	-	-	-	-	-
	AdamW	-	-	-	-	-	-
	Muon	-	-	-	-	-	-
	ISO	-	-	-	-	-	-
Llama-3.2-8B-Instruct	Base	-	-	-	-	-	-
	AdamW	-	-	-	-	-	-
	Muon	-	-	-	-	-	-
	ISO	-	-	-	-	-	-

Table 4: Evaluation results on code benchmarks. We compare ISO against Adam, SGD, and Muon starting from DS-1.5B. ISO results are shaded.

Method	LCB v5 (2024.08.01–2025.02.01)		LCB v6 (2025.02.01–2025.05.01)		Avg.
	avg@8	pass@8	avg@16	pass@16	
DS-1.5B	-	-	-	-	-
Adam	-	-	-	-	-
Muon	-	-	-	-	-
ISO	-	-	-	-	-

7.1 Experimental Results

As shown in Fig. 5, ISO converges substantially faster than AdamW on both Qwen3-1.7B and DS-1.5B, reducing the number of steps needed to reach a target accuracy by up to $\sim 70\%$. We attribute this gain to the geometric constraint enforced by ISO. AdamW operates in unconstrained Euclidean space and may spend early training steps exploring directions that change the spectrum, even though the spectrum is empirically stable under RL post-training. In contrast, ISO fixes Σ and optimizes only the Stiefel factors (U, V), directly restricting updates to the dominant degrees of freedom observed in RL. As a result, ISO follows a more efficient trajectory from the beginning, reaching AdamW’s final performance in far fewer steps and often achieving better final accuracy as well.

7.2 Wall-clock time-to-target.

As shown in Table 5, ISO also improves end-to-end wall-clock time-to-target, not only the number of training steps. This comparison is important because ISO introduces additional structured computation

Table 5: Wall-clock time-to-target comparison on Qwen3-1.7B-Base with DeepMath using 8 NVIDIA A100-SXM4-80GB GPUs. We report both the matched-learning-rate setting and a stronger tuned AdamW baseline.

Setting	Optimizer	Training Steps	Accuracy	Time Cost (s)
Matched LR	AdamW, lr= 1×10^{-6}	200	19.83	35,398
Matched LR	ISO, lr= 1×10^{-6}	60	20.64	13,077
Tuned AdamW	AdamW, lr= 2×10^{-6}	200	20.73	40,101
Tuned AdamW	ISO, lr= 1×10^{-6}	80	20.91	18,370

per step, including the isospectral parameterization and Stiefel retraction. Despite this overhead, ISO reaches the target accuracy in substantially less total training time than AdamW under the same model, dataset, and hardware setup.

We attribute this gain to the same geometric constraint discussed above. AdamW performs cheaper unconstrained updates, but it may require many more rollout-and-update iterations because it explores degrees of freedom that are not central to the RLVR trajectory. In contrast, ISO directly restricts optimization to the Stiefel factors (U, V) while keeping Σ fixed, which improves the amount of useful progress obtained from each sampled batch. Therefore, although ISO is not necessarily cheaper per step, it can be more efficient end-to-end in the time-to-target sense.

7.3 Ablation Study

In this section, we study which Stiefel factor is more important for ISO. For each weight matrix $W \in \mathbb{R}^{m \times n}$, ISO computes a reduced SVD $W = U\Sigma V^\top$, where $U \in \mathbb{R}^{m \times r}$ and $V^\top \in \mathbb{R}^{r \times n}$. Therefore, updating U and updating V do not necessarily introduce the same number of trainable parameters: the parameter count of U is proportional to mr , while the parameter count of $V^\top$ is proportional to rn . This asymmetry becomes important for non-square transformer matrices. For Qwen3-1.7B-Base, the MLP matrices `gate_proj` and `up_proj` have shape (6144, 2048), so their U factors are much larger than their V factors. As a result, the “OnlyU” variant has more trainable parameters than the “OnlyV” variant, as shown in Table 7.

Based on this observation, we design several U/V ablation variants. The `OnlyU` variant updates only the U factor for every matrix, while `OnlyV` updates only the V factor for every matrix. For non-square matrices, `Max+V` updates the larger factor and freezes the smaller factor, whereas `Min+V` updates the smaller factor and freezes the larger factor. For square matrices, both `Max+V` and `Min+V` use V as the tie-breaking factor, hence the “+V” suffix. All variants keep the singular values Σ fixed, so this ablation isolates the effect of choosing different Stiefel degrees of freedom under the same isospectral constraint.

Table 6: Representative matrix shapes in Qwen3-1.7B-Base. Non-square matrices create different trainable parameter counts for the U and V factors.

Matrix	Block	W Shape
<code>q_proj</code>	Self-Attention	(2048, 2048)
<code>k_proj</code>	Self-Attention	(1024, 2048)
<code>v_proj</code>	Self-Attention	(1024, 2048)
<code>o_proj</code>	Self-Attention	(2048, 2048)
<code>gate_proj</code>	MLP	(6144, 2048)
<code>up_proj</code>	MLP	(6144, 2048)
<code>down_proj</code>	MLP	(2048, 6144)

We evaluate these variants on the same math reasoning benchmarks used in the main experiments. This ablation allows us to test whether ISO requires updating both Stiefel factors, or whether most of the improvement can be obtained by updating a subset of the available Stiefel degrees of freedom.

Table 7: Trainable parameter counts of different U/V ablation strategies on Qwen3-1.7B-Base. The percentage is normalized by the largest strategy, Max+V.

Strategy	Trainable Parameters	Relative Ratio
OnlyU	1,116M	82.9%
OnlyV	940M	72.7%
Max+V	1,409M	100.0%
Min+V	646M	55.6%

338 It also controls for the fact that different U/V choices correspond to different trainable parameter
339 counts.

Table 8: Ablation study of U/V update strategies for ISO on math reasoning benchmarks. We report accuracy on five math benchmarks. Results are placeholders until the corresponding experiments finish.

Strategy	AIME24 avg@32	AIME25 avg@32	AMC23 avg@4	Minerva avg@4	Olympia avg@8	Avg.
OnlyU	-	-	-	-	-	-
OnlyV	-	-	-	-	-	-
Max+V	-	-	-	-	-	-
Min+V	-	-	-	-	-	-

340 8 Related Works

341 [zhizhou: Need to discuss with hangqing, do we actually need this section?]

342 [zhizhou:

343 1 Structural update, and manifold update on SFT models.

344 2 Structural update on in post training <https://arxiv.org/pdf/2509.17866>

345]

346 9 Conclusion

347 RLVR is an effective post-training recipe for boosting LLM reasoning, but its adoption is hindered
348 by costly rollouts and low token efficiency under standard optimizers. We show that RLVR in-
349 duces a distinctive geometry: weight updates are nearly isospectral, with singular values largely
350 preserved—unlike the spectral reshaping typical of supervised fine-tuning—while the dominant dy-
351 namics arise from evolution of the singular subspaces, better characterized by ambient transport than
352 by mixing within a fixed subspace. Motivated by this, we propose the **Isospectral Stiefel Optimizer**
353 (**ISO**), which freezes singular values and optimizes only the Stiefel factors via manifold retraction; on
354 reasoning RLVR tasks, ISO matches AdamW’s best performance with up to 70% fewer training steps,
355 suggesting that aligning optimizers with RLVR geometry can substantially improve efficiency and
356 reduce compute/energy costs, and that these geometric insights may guide broader RLVR algorithm
357 design beyond ISO.

358 References

- 359 [1] Chenxin An, Zhihui Xie, Xiaonan Li, Lei Li, Jun Zhang, Shansan Gong, Ming Zhong, Jingjing
360 Xu, Xipeng Qiu, Mingxuan Wang, and Lingpeng Kong. Polaris: A post-training recipe for
361 scaling reinforcement learning on advanced reasoning models, 2025. URL [https://hkunlp.](https://hkunlp.github.io/blog/2025/Polaris)
362 [github.io/blog/2025/Polaris](https://hkunlp.github.io/blog/2025/Polaris).

- [2] Rajendra Bhatia. *Matrix Analysis*. Springer, New York, 1997. ISBN 978-0-387-94846-8.
- [3] Tianzhe Chu, Yuexiang Zhai, Jihan Yang, Shengbang Tong, Saining Xie, Dale Schuurmans, Quoc V Le, Sergey Levine, and Yi Ma. Sft memorizes, rl generalizes: A comparative study of foundation model post-training. *arXiv preprint arXiv:2501.17161*, 2025.
- [4] DeepSeek-AI. Deepseek-rl: Incentivizing reasoning capability in llms via reinforcement learning, 2025. URL <https://arxiv.org/abs/2501.12948>.
- [5] Antonio Andrea Gargiulo, Donato Crisostomi, Maria Sofia Bucarelli, Simone Scardapane, Fabrizio Silvestri, and Emanuele Rodola. Task singular vectors: Reducing task interference in model merging. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 18695–18705, 2025.
- [6] Bingxiang He, Zekai Qu, Zeyuan Liu, Yinghao Chen, Yuxin Zuo, Cheng Qian, Kaiyan Zhang, Weize Chen, Chaojun Xiao, Ganqu Cui, et al. Justrl: Scaling a 1.5 b llm with a simple rl recipe. *arXiv preprint arXiv:2512.16649*, 2025.
- [7] Chaoqun He, Renjie Luo, Yuzhuo Bai, Shengding Hu, Zhen Leng Thai, Junhao Shen, Jinyi Hu, Xu Han, Yujie Huang, Yuxiang Zhang, Jie Liu, Lei Qi, Zhiyuan Liu, and Maosong Sun. Olympiadbench: A challenging benchmark for promoting agi with olympiad-level bilingual multimodal scientific problems, 2024.
- [8] Zhiwei He, Tian Liang, Jiahao Xu, Qiuzhi Liu, Xingyu Chen, Yue Wang, Linfeng Song, Dian Yu, Zhenwen Liang, Wenxuan Wang, Zhuosheng Zhang, Rui Wang, Zhaopeng Tu, Haitao Mi, and Dong Yu. Deepmath-103k: A large-scale, challenging, decontaminated, and verifiable mathematical dataset for advancing reasoning. 2025. URL <https://arxiv.org/abs/2504.11456>.
- [9] Gabriel Ilharco, Marco Tulio Ribeiro, Mitchell Wortsman, Suchin Gururangan, Ludwig Schmidt, Hannaneh Hajishirzi, and Ali Farhadi. Editing models with task arithmetic. *arXiv preprint arXiv:2212.04089*, 2022.
- [10] Aitor Lewkowycz, Anders Andreassen, David Dohan, Ethan Dyer, Henryk Michalewski, Vinay Ramasesh, Ambrose Slone, Cem Anil, Imanol Schlag, Theo Gutman-Solo, et al. Solving quantitative reasoning problems with language models. *Advances in neural information processing systems*, 35:3843–3857, 2022.
- [11] Jiaqi Li, Xiaobo Wang, Wentao Ding, Zihao Wang, Yipeng Kang, Zixia Jia, and Zilong Zheng. Ram: Towards an ever-improving memory system by learning from communications, 2024. URL <https://arxiv.org/abs/2404.12045>.
- [12] Jingyuan Liu, Jianlin Su, Xingcheng Yao, Zhejun Jiang, Guokun Lai, Yulun Du, Yidao Qin, Weixin Xu, Enzhe Lu, Junjie Yan, et al. Muon is scalable for llm training. *arXiv preprint arXiv:2502.16982*, 2025.
- [13] Mingjie Liu, Shizhe Diao, Ximing Lu, Jian Hu, Xin Dong, Yejin Choi, Jan Kautz, and Yi Dong. Prorl: Prolonged reinforcement learning expands reasoning boundaries in large language models. *arXiv preprint arXiv:2505.24864*, 2025.
- [14] Sagnik Mukherjee, Lifan Yuan, Dilek Hakkani-Tur, and Hao Peng. Reinforcement learning finetunes small subnetworks in large language models. *arXiv preprint arXiv:2505.11711*, 2025.
- [15] Yuji Nakatsukasa and Nicholas J. Higham. Backward stability of iterations for computing the polar decomposition. *SIAM Journal on Matrix Analysis and Applications*, 33(2):460–479, 2012. doi: 10.1137/110857544. URL <https://doi.org/10.1137/110857544>.
- [16] Cheng Qian, Emre Can Acikgoz, Qi He, Hongru Wang, Xiushi Chen, Dilek Hakkani-Tür, Gokhan Tur, and Heng Ji. Toolrl: Reward is all tool learning needs. *arXiv preprint arXiv:2504.13958*, 2025.

- [17] Qwen, :, An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, Huan Lin, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiayi Yang, Jingren Zhou, Junyang Lin, Kai Dang, Keming Lu, Keqin Bao, Kexin Yang, Le Yu, Mei Li, Mingfeng Xue, Pei Zhang, Qin Zhu, Rui Men, Runji Lin, Tianhao Li, Tianyi Tang, Tingyu Xia, Xingzhang Ren, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yu Wan, Yuqiong Liu, Zeyu Cui, Zhenru Zhang, and Zihan Qiu. Qwen2.5 technical report, 2025. URL <https://arxiv.org/abs/2412.15115>.
- [18] John Schulman and Thinking Machines Lab. Lora without regret. *Thinking Machines Lab: Connectionism*, 2025. doi: 10.64434/tml.20250929. <https://thinkingmachines.ai/blog/lora/>.
- [19] Idan Shenfeld, Jyothish Pari, and Pulkit Agrawal. RL’s razor: Why online reinforcement learning forgets less. *arXiv preprint arXiv:2509.04259*, 2025.
- [20] Guangming Sheng, Chi Zhang, Zilingfeng Ye, Xibin Wu, Wang Zhang, Ru Zhang, Yanghua Peng, Haibin Lin, and Chuan Wu. Hybridflow: A flexible and efficient rlhf framework. In *Proceedings of the Twentieth European Conference on Computer Systems*, pages 1279–1297, 2025.
- [21] Qwen Team. Qwen3 technical report, 2025. URL <https://arxiv.org/abs/2505.09388>.
- [22] Jiakang Wang, Runze Liu, Lei Lin, Wenping Hu, Xiu Li, Fuzheng Zhang, Guorui Zhou, and Kun Gai. Aspo: Asymmetric importance sampling policy optimization. *arXiv preprint arXiv:2510.06062*, 2025.
- [23] Yinjie Wang, Ling Yang, Ye Tian, Ke Shen, and Mengdi Wang. Co-evolving llm coder and unit tester via reinforcement learning. *arXiv preprint arXiv:2506.03136*, 2025.
- [24] Per-Åke Wedin. Perturbation bounds in connection with singular value decomposition. *BIT Numerical Mathematics*, 12(1):99–111, 1972.
- [25] Kaiyue Wen, David Hall, Tengyu Ma, and Percy Liang. Fantastic pretraining optimizers and where to find them. *arXiv preprint arXiv:2509.02046*, 2025.
- [26] xAI. Grok: Ai assistant, 2025. URL <https://x.ai/grok>. Accessed: 2025-09-24, continuously updated.
- [27] Prateek Yadav, Derek Tam, Leshem Choshen, Colin A Raffel, and Mohit Bansal. Ties-merging: Resolving interference when merging models. *Advances in neural information processing systems*, 36:7093–7115, 2023.
- [28] Hongli Yu, Tinghong Chen, Jiangtao Feng, Jiangjie Chen, Weinan Dai, Qiyang Yu, Ya-Qin Zhang, Wei-Ying Ma, Jingjing Liu, Mingxuan Wang, et al. Memagent: Reshaping long-context llm with multi-conv rl-based memory agent. *arXiv preprint arXiv:2507.02259*, 2025.
- [29] Yifan Zhang and Team Math-AI. American invitational mathematics examination (aime) 2024, 2024.
- [30] Yifan Zhang and Team Math-AI. American invitational mathematics examination (aime) 2025, 2025.
- [31] Yushun Zhang, Congliang Chen, Ziniu Li, Tian Ding, Chenwei Wu, Diederik P Kingma, Yinyu Ye, Zhi-Quan Luo, and Ruoyu Sun. Adam-mini: Use fewer learning rates to gain more. *arXiv preprint arXiv:2406.16793*, 2024.
- [32] Hanqing Zhu, Zhenyu Zhang, Wenyan Cong, Xi Liu, Sem Park, Vikas Chandra, Bo Long, David Z Pan, Zhangyang Wang, and Jinwon Lee. Apollo: Sgd-like memory, adamw-level performance. *Proceedings of Machine Learning and Systems*, 7, 2025.
- [33] Hanqing Zhu, Zhenyu Zhang, Hanxian Huang, DiJia Su, Zechun Liu, Jiawei Zhao, Igor Fedorov, Hamed Pirsiavash, Zhizhou Sha, Jinwon Lee, et al. The path not taken: RLvr provably learns off the principals. *arXiv preprint arXiv:2511.08567*, 2025.
- [34] Zilin Zhu, Chengxing Xie, Xin Lv, and slime Contributors. slime: An llm post-training framework for rl scaling. <https://github.com/THUDM/slime>, 2025. GitHub repository. Corresponding author: Xin Lv.

Appendix

458

.1 Training-Free Merging Baselines

460 We first review the training-free baselines used for comparison: Task Arithmetic, TIES-Merging,
461 TSV-Merge, and RAM. All four methods start from the same Euclidean weight-space representation.
462 Let θ_{pre} denote the shared pretrained model and let $\{\theta_i\}_{i=1}^T$ be T expert models fine-tuned from this
463 same initialization. The task vector of expert i is

$$\tau_i = \theta_i - \theta_{\text{pre}}. \quad (15)$$

464 A data-free merging method constructs a merged update τ_{merge} without using training data and
465 outputs

$$\theta_{\text{merge}} = \theta_{\text{pre}} + \tau_{\text{merge}}. \quad (16)$$

466 For a matrix-shaped parameter in layer ℓ , we use the layer-wise notation

$$\Delta_i^{(\ell)} = W_i^{(\ell)} - W_{\text{pre}}^{(\ell)}, \quad W^{(\ell)} \in \mathbb{R}^{m_\ell \times n_\ell}. \quad (17)$$

.1.1 Task Arithmetic

468 **Idea.** Task Arithmetic merges experts by linearly combining their task vectors and adding the
469 resulting update back to the pretrained model:

$$\tau_{\text{TA}} = \lambda \sum_{i=1}^T \tau_i, \quad \theta_{\text{TA}} = \theta_{\text{pre}} + \tau_{\text{TA}}. \quad (18)$$

470 Equivalently, a weighted variant uses

$$\tau_{\text{TA}} = \sum_{i=1}^T \lambda_i \tau_i. \quad (19)$$

471 Setting $\lambda_i = 1/T$ recovers task-vector averaging, while setting $\lambda_i = \lambda$ recovers the common
472 “sum-then-scale” form.

473 **Properties.** Task Arithmetic is computationally cheap, requiring only vector additions and a global
474 scaling coefficient. However, it does not explicitly address sign conflicts, redundant updates, layer-
475 wise geometry, or sparse task-specific signals. As a result, interference and signal dilution can occur
476 when expert updates overlap or when important updates are distributed over disjoint parameter
477 regions.

.1.2 TIES-Merging

479 **Idea.** TIES-Merging follows three steps: *Trim*, *Elect Sign*, and *Merge* [?]. It first removes small-
480 magnitude entries from each task vector, then elects a consensus sign for every coordinate, and finally
481 averages only the updates whose signs agree with the elected sign.

482 Let $\rho \in (0, 1]$ be the retained density. The trimmed update for expert i is

$$m_i = \text{TopKMask}(|\tau_i|, \rho), \quad \tilde{\tau}_i = m_i \odot \tau_i, \quad (20)$$

483 where $m_i^p = 1$ indicates that coordinate p is among the top- ρ largest entries of $|\tau_i|$. The elected sign
484 at coordinate p is

$$\gamma^p = \text{sign} \left(\sum_{i=1}^T \tilde{\tau}_i^p \right). \quad (21)$$

485 TIES then keeps only the experts aligned with this sign:

$$A_p = \{i : \tilde{\tau}_i^p \neq 0, \text{sign}(\tilde{\tau}_i^p) = \gamma^p\}. \quad (22)$$

486 The merged coordinate is

$$\tau_{\text{TIES}}^p = \begin{cases} \frac{1}{|A_p|} \sum_{i \in A_p} \tilde{\tau}_i^p, & |A_p| > 0, \\ 0, & |A_p| = 0, \end{cases} \quad (23)$$

487 and the final model is

$$\theta_{\text{TIES}} = \theta_{\text{pre}} + \lambda \tau_{\text{TIES}}. \quad (24)$$

488 **Properties.** TIES remains an element-wise method, but it is more robust than naive averaging when
 489 many small updates are irrelevant or when experts update the same coordinate in opposite directions.
 490 Its main limitation is that it does not explicitly model matrix-level subspace geometry.

491 .1.3 TSV-Merge

492 **Idea.** TSV-Merge, short for Task Singular Vectors Merge, operates on layer-wise task matrices rather
 493 than fully flattened vectors [?]. For each 2D weight matrix, TSV decomposes each expert update
 494 with SVD, keeps the leading singular components, orthogonalizes the concatenated singular-vector
 495 bases, and reconstructs the merged matrix.

496 For layer ℓ , compute

$$\Delta_i^{(\ell)} = U_i \Sigma_i V_i^\top. \quad (25)$$

497 After truncating to rank k_ℓ ,

$$\Delta_i^{(\ell)} \approx U_i^{(k)} \Sigma_i^{(k)} V_i^{(k)\top}. \quad (26)$$

498 The truncated bases and spectra are concatenated as

$$U = [U_1^{(k)} | \dots | U_T^{(k)}], \quad V = [V_1^{(k)} | \dots | V_T^{(k)}], \quad \Sigma = \text{blockdiag}(\Sigma_1^{(k)}, \dots, \Sigma_T^{(k)}). \quad (27)$$

499 TSV reduces singular-vector interference by orthogonalizing U and V . In the whitening form,

$$U_\perp = U(U^\top U)^{-1/2}, \quad V_\perp = V(V^\top V)^{-1/2}. \quad (28)$$

500 A numerically stable implementation can use the orthogonal Procrustes form. If $U = P_U D_U Q_U^\top$ and
 501 $V = P_V D_V Q_V^\top$, then

$$U_\perp = P_U Q_U^\top, \quad V_\perp = P_V Q_V^\top. \quad (29)$$

502 The merged task matrix is reconstructed as

$$\hat{\Delta}^{(\ell)} = U_\perp \Sigma V_\perp^\top, \quad W_{\text{TSV}}^{(\ell)} = W_{\text{pre}}^{(\ell)} + \alpha \hat{\Delta}^{(\ell)}. \quad (30)$$

503 For non-matrix parameters, such as biases or normalization vectors, TSV typically falls back to Task
 504 Arithmetic.

505 **Properties.** TSV is a spectral, subspace-level method. By decorrelating singular-vector bases, it can
 506 reduce interference that is invisible to element-wise rules. This comes at a higher computational cost
 507 because SVDs are required for matrix-shaped layers. The rank k_ℓ is either treated as a hyperparameter
 508 or chosen through an automatic rank-reduction rule.

509 .1.4 RAM

510 **Idea.** RAM is designed for sparse RL updates, where naive averaging can dilute coordinates that are
 511 important for only one expert. For each coordinate p , RAM identifies the experts with non-negligible
 512 updates:

$$A_p = \{i \in \{1, \dots, T\} : |\tau_i^p| > \epsilon_{\text{act}}\}, \quad c^p = |A_p|, \quad (31)$$

513 where ϵ_{act} is an active-update threshold. The merged coordinate is

$$\tau_{\text{RAM}}^p = \begin{cases} 0, & c^p = 0, \\ \tau_j^p, & c^p = 1 \text{ and } A_p = \{j\}, \\ \frac{1}{c^p} \sum_{i \in A_p} \tau_i^p, & c^p \geq 2. \end{cases} \quad (32)$$

514 Thus, inactive coordinates are discarded, task-unique coordinates are preserved without being divided
 515 by T , and shared coordinates are averaged only over the experts that actively update them. The final
 516 model is

$$\theta_{\text{RAM}} = \theta_{\text{pre}} + \tau_{\text{RAM}}. \quad (33)$$

517 **Properties.** RAM is element-wise like TIES, but its selection criterion is different. Instead of
 518 trimming by global magnitude density and resolving signs, RAM separates inactive, unique, and
 519 shared update regions. This makes it suitable when RL-trained experts store task-specific behavior in
 520 sparse and partially non-overlapping parameter subsets.

Table 9: Core differences among the training-free merging baselines.

Method	Main operation	Main issue addressed
Task Arithmetic	Linear sum or average	Basic data-free model merging
TIES-Merging	Trim + sign election + disjoint mean	Redundant updates and sign conflicts
TSV-Merge	SVD truncation + basis orthogonalization	Singular-vector subspace interference
RAM	Active-coordinate averaging; unique-coordinate preservation	Signal dilution in sparse RL updates

1.5 Comparison

A Isospectral dynamics under sublinear cumulative updates

We model reinforcement-learning post-training as an approximately isospectral process: the spectrum is largely preserved while the singular subspaces rotate. This viewpoint is supported by Theorem A.1, which shows that under a per-step update bound and sublinear cumulative updates, the singular values Σ_t exhibit only sublinear drift from the initial spectrum Σ_0 . As a result, the dominant degrees of freedom in W_t can be attributed to the evolution of (U_t, V_t) rather than substantial changes in Σ_t .

Theorem A.1 (Isospectral dynamics under sublinear cumulative updates, formal version of Theorem ??). *Let $\{W_t\}_{t \geq 0} \subset \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ be a sequence of weight matrices generated by reinforcement-learning post-training. For each t , let $W_t = U_t \Sigma_t V_t^\top$ be a singular value decomposition, where $\Sigma_t \in \mathbb{R}^{r \times r}$ is diagonal with nonnegative entries and $r = \text{rank}(W_t)$. Assume the successive updates satisfy $\|W_{t+1} - W_t\|_F \leq \epsilon_t$, and the cumulative update size is sublinear: $\sum_{t=0}^{T-1} \epsilon_t = o(T)$ as $T \rightarrow \infty$. If $\sigma_{\min}(\Sigma_0) > 0$, then the singular-value drift relative to the initial spectrum is sublinear. In particular, there exists a constant $C > 0$ such that for all $t \geq 1$,*

$$\frac{\|\Sigma_t - \Sigma_0\|_F}{\|\Sigma_0\|_F} \leq \frac{C \sum_{k=0}^{t-1} \epsilon_k}{\sigma_{\min}(\Sigma_0)}. \quad (34)$$

Consequently, if $\sum_{k=0}^{t-1} \epsilon_k = o(t)$, then

$$\frac{\|\Sigma_t - \Sigma_0\|_F}{\|\Sigma_0\|_F} = o(1) \quad \text{as } t \rightarrow \infty. \quad (35)$$

The proof is provided in A.1. By Theorem A.1, $\|\Sigma_t - \Sigma_0\|_F / \|\Sigma_0\|_F$ is controlled by the cumulative update size, implying that spectral drift remains small when updates accumulate sublinearly. This provides the theoretical basis for fixing (or tightly constraining) Σ and focusing our analysis on orthogonality-preserving updates of U_t and V_t on Stiefel manifolds.

A.1 Proof of Theorem A.1

Proof. Fix $t \geq 0$ and define the update $\Delta_t := W_{t+1} - W_t$, so that $W_{t+1} = W_t + \Delta_t$. By Weyl's inequality [2] for singular values, for any $A, B \in \mathbb{R}^{m \times n}$ and any index i ,

$$|\sigma_i(A + B) - \sigma_i(A)| \leq \|B\|_2. \quad (36)$$

Applying (36) with $A = W_t$ and $B = \Delta_t$ yields

$$|\sigma_{i,t+1} - \sigma_{i,t}| = |\sigma_i(W_t + \Delta_t) - \sigma_i(W_t)| \leq \|\Delta_t\|_2. \quad (37)$$

Using $\|\Delta_t\|_2 \leq \|\Delta_t\|_F$ and the assumption $\|W_{t+1} - W_t\|_F = \|\Delta_t\|_F \leq \epsilon_t$, we obtain the per-step bound

$$|\sigma_{i,t+1} - \sigma_{i,t}| \leq \epsilon_t. \quad (38)$$

Summing (38) over $k = 0, \dots, t-1$ and using the triangle inequality gives, for each i ,

$$|\sigma_{i,t} - \sigma_{i,0}| \leq \sum_{k=0}^{t-1} |\sigma_{i,k+1} - \sigma_{i,k}| \leq \sum_{k=0}^{t-1} \epsilon_k. \quad (39)$$

Let r denote the number of singular values in the thin SVD of W_t (in particular $r \leq \min\{d_{\text{out}}, d_{\text{in}}\}$).
Then

$$\begin{aligned}\|\Sigma_t - \Sigma_0\|_F^2 &= \sum_{i=1}^r (\sigma_{i,t} - \sigma_{i,0})^2 \\ &\leq \sum_{i=1}^r \left(\sum_{k=0}^{t-1} \epsilon_k \right)^2 = r \left(\sum_{k=0}^{t-1} \epsilon_k \right)^2,\end{aligned}\tag{40}$$

and hence

$$\|\Sigma_t - \Sigma_0\|_F \leq \sqrt{r} \sum_{k=0}^{t-1} \epsilon_k.\tag{41}$$

Moreover, since Σ_0 is diagonal with nonnegative entries and $\sigma_{\min}(\Sigma_0) > 0$,

$$\begin{aligned}\|\Sigma_0\|_F &= \left(\sum_{i=1}^r \sigma_{i,0}^2 \right)^{1/2} \geq (r \sigma_{\min}(\Sigma_0)^2)^{1/2} \\ &= \sqrt{r} \sigma_{\min}(\Sigma_0).\end{aligned}\tag{42}$$

Combining (41) and (42) yields the relative bound

$$\frac{\|\Sigma_t - \Sigma_0\|_F}{\|\Sigma_0\|_F} \leq \frac{\sqrt{r} \sum_{k=0}^{t-1} \epsilon_k}{\sqrt{r} \sigma_{\min}(\Sigma_0)} = \frac{\sum_{k=0}^{t-1} \epsilon_k}{\sigma_{\min}(\Sigma_0)}.\tag{43}$$

Finally, if $\sum_{k=0}^{t-1} \epsilon_k = o(t)$, then the right-hand side of (43) is $o(1)$ as $t \rightarrow \infty$, which proves the claim. $\square$

Beyond spectral stability, a central question in reinforcement-learning post-training is whether the dominant representation subspaces drift in a controlled manner. Theorem A.2 addresses this by quantifying the stability of the top- k left-singular subspace when the dynamics are (approximately) isospectral and the initial spectrum has exponential decay. Under bounded step updates, the theorem gives an explicit operator-norm bound on the deviation of the associated projector P_t , relating cumulative subspace drift to the cumulative update size and the spectral gap implied by exponential decay.

A.2 Subspace stability under isospectral dynamics

Theorem A.2 (Subspace stability under isospectral dynamics). *Let $\{W_t\}_{t \geq 0} \subset \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ be a sequence of matrices with thin singular value decompositions $W_t = U_t \Sigma_t V_t^\top$. Assume: Isospectrality: $\Sigma_t = \Sigma_0$ for all $t \geq 0$, Exponential spectral decay: there exist constants $C > 0$ and $a > 0$ such that, for all $i \geq 1$, $\sigma_i(\Sigma_0) = C e^{-ai}$. and Bounded update steps: $\|W_{t+1} - W_t\|_2 \leq \epsilon_t$ for all $t \geq 0$. Fix an integer $k \geq 1$, and define the orthogonal projector onto the span of the top- k left singular vectors of W_t by*

$$P_t := U_{t,1:k} U_{t,1:k}^\top,\tag{44}$$

where $U_{t,1:k} \in \mathbb{R}^{d_{\text{out}} \times k}$ contains the first k columns of U_t . Then, for any $T \geq 1$,

$$\|P_T - P_0\|_2 \leq \frac{e^{ak}}{C(1 - e^{-a})} \sum_{t=0}^{T-1} \epsilon_t.\tag{45}$$

Theorem A.2 shows that the cumulative change of the top- k projector satisfies Equation (45), so subspace drift remains small when updates accumulate slowly. The prefactor highlights the role of spectral decay: stronger decay (larger a) yields a larger effective gap among leading modes, which tightens the stability control for fixed k . The detailed proof is provided in Section A.3

573 A.3 Proof of Theorem A.2

574 *Proof.* Under the exponential-decay assumption in Theorem A.2, the spectral gap between the k -th
575 and $(k + 1)$ -th singular values of Σ_0 is

$$\begin{aligned}\delta_k &:= \sigma_k(\Sigma_0) - \sigma_{k+1}(\Sigma_0) \\ &= Ce^{-ak} - Ce^{-a(k+1)} \\ &= Ce^{-ak}(1 - e^{-a}).\end{aligned}\tag{46}$$

576 By the isospectrality assumption $\Sigma_t = \Sigma_0$ for all t , the gap δ_k is time-invariant.

577 Fix $t \geq 0$. Let $\mathcal{U}_t := \text{span}(U_{t,1:k})$ and $\mathcal{U}_{t+1} := \text{span}(U_{t+1,1:k})$ denote the top- k left-singular
578 subspaces of W_t and W_{t+1} , respectively, and let $P_t = U_{t,1:k}U_{t,1:k}^\top$ be the corresponding orthogonal
579 projector. By Wedin's $\sin \Theta$ theorem [24], if $\|W_{t+1} - W_t\|_2 \leq \epsilon_t$, then

$$\|\sin \Theta(\mathcal{U}_t, \mathcal{U}_{t+1})\|_2 \leq \frac{\|W_{t+1} - W_t\|_2}{\delta_k} \leq \frac{\epsilon_t}{\delta_k}.\tag{47}$$

580 Moreover, for orthogonal projectors onto $\mathcal{U}_t$ and $\mathcal{U}_{t+1}$, the standard identity relates principal angles
581 to projector differences:

$$\|P_{t+1} - P_t\|_2 = \|\sin \Theta(\mathcal{U}_t, \mathcal{U}_{t+1})\|_2.\tag{48}$$

582 Combining (47) and (48) yields the per-step bound

$$\|P_{t+1} - P_t\|_2 \leq \frac{\epsilon_t}{\delta_k}.\tag{49}$$

583 To control the cumulative drift up to step T , apply the triangle inequality and sum (49) over $t =$
584 $0, \dots, T - 1$:

$$\|P_T - P_0\|_2 \leq \sum_{t=0}^{T-1} \|P_{t+1} - P_t\|_2 \leq \sum_{t=0}^{T-1} \frac{\epsilon_t}{\delta_k}.\tag{50}$$

585 Substituting (46) into (50) gives

$$\begin{aligned}\|P_T - P_0\|_2 &\leq \frac{1}{Ce^{-ak}(1 - e^{-a})} \sum_{t=0}^{T-1} \epsilon_t \\ &= \frac{e^{ak}}{C(1 - e^{-a})} \sum_{t=0}^{T-1} \epsilon_t,\end{aligned}\tag{51}$$

586 which is the claimed bound. $\square$

587 B Subspace Rotation and Ambient Transport

588 B.1 Left and Right Multiplication

589 **Right-multiplication** by an orthogonal matrix $R \in O(r)$,

$$U \mapsto UR, \quad V \mapsto VR,\tag{52}$$

590 where $U \in \mathbb{R}^{n \times r}$, $V \in \mathbb{R}^{m \times r}$, and $R \in \mathbb{R}^{r \times r}$. This is equivalent to stating that RL primarily performs
591 a rotation in the **output feature space**. This transformation mixes the *columns* of U and V through
592 an r -dimensional orthogonal rotation, while leaving their column spaces unchanged. In particular,

$$\text{span}(UR) = \text{span}(U) \subset \mathbb{R}^n, \quad \text{span}(VR) = \text{span}(V) \subset \mathbb{R}^m,\tag{53}$$

593 so the associated left and right singular subspaces are preserved. Geometrically, right-multiplication
594 corresponds to a change of orthonormal basis within the same r -dimensional subspace, acting only
595 on the intrinsic coordinate system of the columns. This operation therefore represents **inner mixing**
596 on the Stiefel manifold.

597 **Left-multiplication** by orthogonal transformations $Q_U \in O(n)$ and $Q_V \in O(m)$,

$$U \mapsto Q_U U, \quad V \mapsto Q_V V,\tag{54}$$

where $U \in \mathbb{R}^{n \times r}$, $V \in \mathbb{R}^{m \times r}$, $Q_U \in \mathbb{R}^{n \times n}$, and $Q_V \in \mathbb{R}^{m \times m}$. This is equivalent to stating that RL primarily performs a rotation in the **input feature space**. This transformation applies ambient-space rotations to the columns of U and V , thereby transporting their r -dimensional column spaces within $\mathbb{R}^n$ and $\mathbb{R}^m$, respectively. Although column orthonormality is preserved, the resulting subspaces generally satisfy

$$\text{span}(Q_U U) \neq \text{span}(U) \subset \mathbb{R}^n, \quad \text{span}(Q_V V) \neq \text{span}(V) \subset \mathbb{R}^m. \quad (55)$$

Geometrically, left-multiplication corresponds to motion of the associated Grassmann points on $\text{Gr}(n, r)$ and $\text{Gr}(m, r)$, representing **ambient transport** of the low-rank singular subspaces rather than a mere change of basis.

B.2 Subspace Inner Mixing

We now state a formal version of Theorem ??, which quantifies how small operator-norm updates constrain the drift of the leading singular subspace under isospectral dynamics with an exponentially decaying spectrum.

Assumption B.1 (Isospectral dynamics with exponential spectrum). Let $\{W_t\}_{t \geq 0} \subset \mathbb{R}^{n \times m}$ admit the SVD $W_t = U_t \Sigma_t V_t^\top$. Assume:

1. **Isospectrality.** The singular values are invariant along training,

$$\Sigma_t = \Sigma_0, \quad \text{equivalently } \sigma_i(W_t) = \sigma_i(W_0) \quad \forall i, t. \quad (56)$$

2. **Exponential decay at initialization.** There exist constants $c_1, c_2 > 0$ and $\alpha > 0$ such that

$$c_1 e^{-\alpha i} \leq \sigma_i(W_0) \leq c_2 e^{-\alpha i}, \quad i = 1, 2, \dots \quad (57)$$

3. **Small operator-norm updates.** The dynamics satisfy

$$W_{t+1} = W_t + \Delta_t, \quad \|\Delta_t\|_2 \leq \varepsilon_t. \quad (58)$$

Theorem B.2 (Controlled Subspace Inner Mixing, formal version of Theorem ??). Let $P_t := U_{t,1:k} U_{t,1:k}^\top$ denote the orthogonal projector onto the leading k -dimensional left singular subspace of W_t . Under Assumption B.1 and assuming the exact exponential spectrum $\sigma_i(W_0) = C e^{-\alpha i}$, the cumulative subspace deviation satisfies

$$\|P_T - P_0\|_2 \leq \frac{e^{\alpha k}}{C(1 - e^{-\alpha})} \sum_{t=0}^{T-1} \varepsilon_t. \quad (59)$$

Proof. The proof follows directly from that of Theorem A.2 in Section A.3, with only minor notational changes. $\square$

What “spectrum invariant” guarantees about the subspace. Because $\Sigma_t = \Sigma_0$, the ordering of singular values is fixed and no eigenvalue crossing can occur; hence the *same indexed* leading- k spectral subspace (the one associated with $\{\sigma_1, \dots, \sigma_k\}$) remains identifiable throughout RL. Exponential decay then yields a concrete, persistent eigengap δ , and the bounds above show that the leading subspace can only move by a controlled amount; in particular, if $\sum_{t=0}^{T-1} \varepsilon_t = o(\delta)$, then $\|P_T - P_0\|_2 = o(1)$, i.e., the leading singular subspace is asymptotically invariant.

B.3 Ambient Transport

We now state a formal version of Theorem ?. To obtain a nontrivial lower bound on subspace motion, additional assumptions are required to ensure persistent cross-subspace coupling and to rule out cancellation across steps.

Assumption B.3 (Cross-subspace coupling and persistence). Let $\mathcal{U}_t := \text{span}(U_{t,1:k})$ and let $U_{t,k+1:}$ denote an orthonormal basis of $\mathcal{U}_t^\perp$. Define the cross-subspace coupling magnitude

$$\eta_t := \|U_{t,k+1:}^\top \Delta_t U_{t,1:k}\|_2. \quad (60)$$

633 Assume there exists a known sequence $\{\eta_t\}_{t \geq 0}$ such that

$$\eta_t \geq \underline{\eta}_t \quad \text{for all } t. \quad (61)$$

634 Moreover, assume a no-cancellation (directional persistence) condition: there exists $\gamma \in (0, 1]$ such
635 that

$$\begin{aligned} \|P_T - P_0\|_2 &\geq \gamma \sum_{t=0}^{T-1} \|P_{t+1} - P_t\|_2, \\ P_t &:= U_{t,1:k} U_{t,1:k}^\top. \end{aligned} \quad (62)$$

636 **Theorem B.4** (Ambient Transport, formal version of Theorem ??). *Under Assumptions B.1 and B.3,*
637 *and assuming the exact exponential spectrum $\sigma_i(W_0) = Ce^{-\alpha i}$, the cumulative deviation of the*
638 *leading k -dimensional left singular subspace satisfies*

$$\|P_T - P_0\|_2 \geq \gamma \sum_{t=0}^{T-1} \frac{\eta_t}{Ce^{-\alpha k}(1 - e^{-\alpha}) + \varepsilon_t}. \quad (63)$$

639 *Proof.* Fix t and recall from Assumption B.1(iii) that $W_{t+1} = W_t + \Delta_t$ with $\|\Delta_t\|_2 \leq \varepsilon_t$. Let
640 $\delta := \sigma_k(W_t) - \sigma_{k+1}(W_t)$. By Assumption B.1(i), δ is time-invariant and equals $\sigma_k(W_0) - \sigma_{k+1}(W_0)$.
641 A two-sided singular-subspace perturbation bound (Wedin-type $\sin \Theta$ inequality) yields

$$\begin{aligned} \|\sin \Theta(\mathcal{U}_t, \mathcal{U}_{t+1})\|_2 &\geq \frac{\|U_{t,k+1}^\top \Delta_t U_{t,1:k}\|_2}{\delta + \|\Delta_t\|_2} \\ &= \frac{\eta_t}{\delta + \|\Delta_t\|_2}. \end{aligned} \quad (64)$$

642 By Assumption B.3, $\eta_t \geq \underline{\eta}_t$, and by Assumption B.1(iii), $\|\Delta_t\|_2 \leq \varepsilon_t$, hence

$$\|\sin \Theta(\mathcal{U}_t, \mathcal{U}_{t+1})\|_2 \geq \frac{\eta_t}{\delta + \varepsilon_t}. \quad (65)$$

643 By Assumption B.1(ii),

$$\begin{aligned} \delta &= \sigma_k(W_0) - \sigma_{k+1}(W_0) \\ &\geq c_1 e^{-\alpha k} - c_2 e^{-\alpha(k+1)} \\ &= e^{-\alpha k} (c_1 - c_2 e^{-\alpha}). \end{aligned} \quad (66)$$

644 Substituting this lower bound into Equation (65) gives

$$\|\sin \Theta(\mathcal{U}_t, \mathcal{U}_{t+1})\|_2 \geq \frac{\eta_t}{e^{-\alpha k} (c_1 - c_2 e^{-\alpha}) + \varepsilon_t}. \quad (67)$$

645 Recall the projector identity aligned with Equation (48)

$$\begin{aligned} \|P_{t+1} - P_t\|_2 &= \|\sin \Theta(\mathcal{U}_t, \mathcal{U}_{t+1})\|_2, \\ P_t &:= U_{t,1:k} U_{t,1:k}^\top. \end{aligned} \quad (68)$$

646 Under the persistence condition in Assumption B.3,

$$\begin{aligned} \|P_T - P_0\|_2 &\geq \gamma \sum_{t=0}^{T-1} \|P_{t+1} - P_t\|_2 \\ &= \gamma \sum_{t=0}^{T-1} \|\sin \Theta(\mathcal{U}_t, \mathcal{U}_{t+1})\|_2 \\ &\geq \gamma \sum_{t=0}^{T-1} \frac{\eta_t}{\delta + \varepsilon_t}. \end{aligned} \quad (69)$$

647 Finally, substituting the explicit lower bound on δ from Equation (66) gives

$$\|P_T - P_0\|_2 \geq \gamma \sum_{t=0}^{T-1} \frac{\eta_t}{Ce^{-\alpha k}(1 - e^{-\alpha}) + \varepsilon_t}. \quad (70)$$

648 $\square$

Remarks. First, without a coupling lower bound (such as $\eta_t \geq \underline{\eta}_t$), no nontrivial lower bound exists because the update can stay entirely within $\mathcal{U}_t$ and leave P_t unchanged. Second, without a persistence/no-cancellation condition, the global motion $\|P_T - P_0\|_2$ can be much smaller than the sum of per-step motions because successive rotations may cancel.

C Newton–Schulz Iteration Analysis

Lemma C.1 (Finite-precision Newton–Schulz polar factor error bound with condition number). [15] Let $A \in \mathbb{C}^{s \times n}$ with $s \geq n$ and $\text{rank}(A) = n$, and let $A = U_A H$ be its (thin) polar decomposition, where $U_A^* U_A = I_n$ and $H = (A^* A)^{1/2} > 0$. Consider the (cubic) Newton–Schulz iteration

$$X_{k+1} = X_k P(X_k^* X_k), \quad X_0 = A/\alpha, \quad P(S) := \frac{15}{8}I - \frac{5}{4}S + \frac{3}{8}S^2, \quad (71)$$

and denote by $\hat{X}_k$ the iterates computed in floating-point arithmetic with unit roundoff u .

Then, the forward error of the computed polar factor is bounded by the condition number:

$$\|\hat{U} - U_A\|_2 \leq c_2 u \kappa_2(A) + O(u \delta \kappa_2(A)) + O(u^2), \quad \kappa_2(A) := \|A\|_2 \|A^\dagger\|_2. \quad (72)$$

where $A^\dagger$ denotes the Moore–Penrose pseudoinverse and $c_1, c_2 \geq 1$ are modest constants independent of u .

This shows that Newton–Schulz iteration finds the orthonormal factor of a matrix in a numerically stable way in finite-precision arithmetic. The error in this factor is proportional to the precision and the condition number of our input matrix. When the matrix is well-conditioned, our resulting orthonormal factor is very close to the exact one. This lemma guarantees that replacing an exact SVD-based projection with a Newton–Schulz-based method does not add much error, and all deviations are controlled and stable.

D Precision Issues in PyTorch SVD Decomposition.

Our goal is to keep the singular spectrum of each weight matrix unchanged throughout training. A subtle but important practical issue is that the singular values can be highly sensitive to numerical errors introduced by the SVD routine and subsequent reconstruction.

In PyTorch, SVD can be executed under four common configurations, determined by numerical precision and hardware: (i) fp32 on CPU, (ii) fp64 on CPU, (iii) fp32 on GPU, and (iv) fp64 on GPU. To quantify the trade-off between accuracy and runtime, we run the following sanity check. We conduct experiments on Qwen3 1.7B base model.

1. We extract the q project layer in the 10-th layer of Qwen3 1.7B base model $W_0 \in \mathbb{R}^{2048 \times 2048}$.
2. We compute $W_0 = U \Sigma V^\top$ under each configuration.
3. We reconstruct $W_1 = U \Sigma V^\top$.
4. We measure (a) the mean MSE loss of singular values between W_1 and W_0 , and (b) the wall-clock time.

Table 10: Runtime and singular-value drift of PyTorch SVD under different precision and device configurations.

Precision	Device	Singular-value drift	Time (s)
fp32	CPU	3.6633e-7	2.115
fp64	CPU	1.1126e-8	3.699
fp32	GPU	7.7899e-4	0.894
fp64	GPU	2.8924e-8	0.923

Table 10 summarizes the results. We find that the CPU-based variants are prohibitively slow and thus unsuitable for our training pipeline.

More importantly, the combination of fp32 + GPU introduces unexpectedly large spectral distortion: the singular-value drift after SVD+reconstruction can reach 7.7899e-4, which is orders of magnitude larger than the typical singular-value variation observed during standard RLVR training (1.988e-5 in Qwen3 1.7B base trained with AdamW with learning rate 1e-6). For these reasons, we adopt fp64 SVD on GPU in our implementation to meet the numerical requirements of $\text{msign}(\cdot)$.

E Convergence analysis

We study the optimization of W under a fixed singular-value spectrum, i.e., $W = U\Sigma V^\top$ with Σ fixed and (U, V) updated by gradient descent.

Theorem E.1 (Monotone convergence for Algorithm ??). *Assume f is bounded below and has L_f -Lipschitz Euclidean gradient: $\|\nabla f(X) - \nabla f(Y)\|_F \leq L_f\|X - Y\|_F$ for all X, Y . Let $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_r) > 0$ be fixed. Define $F(U, V) = f(U\Sigma V^\top)$ on $\mathcal{M} := \text{St}(m, r) \times \text{St}(n, r)$ endowed with the canonical metric, and denote the Riemannian gradient by $\text{grad}F$.*

Consider the retraction-based Riemannian gradient descent update with polar retractions

$$U_{t+1} = \text{msign}(U_t - \eta \text{grad}_U F(U_t, V_t)), \quad V_{t+1} = \text{msign}(V_t - \eta \text{grad}_V F(U_t, V_t)). \quad (73)$$

Then there exists a constant $L_F > 0$ such that for any $\eta \in (0, 1/L_F]$,

$$F(U_{t+1}, V_{t+1}) \leq F(U_t, V_t) - \frac{\eta}{2} \left(\|\text{grad}_U F(U_t, V_t)\|_F^2 + \|\text{grad}_V F(U_t, V_t)\|_F^2 \right). \quad (74)$$

Consequently $F(U_t, V_t)$ is non-increasing and convergent, and

$$\sum_{t=0}^{\infty} \|\text{grad}F(U_t, V_t)\|_F^2 < \infty, \quad \lim_{t \rightarrow \infty} \|\text{grad}F(U_t, V_t)\|_F = 0. \quad (75)$$

Hence every accumulation point $(U_\star, V_\star)$ is a first-order stationary point of F on $\mathcal{M}$.

The proof is provided in Section E.1. The orthogonality constraints $U^\top U = I$ and $V^\top V = I$ are enforced by the polar map msign , which acts as a retraction onto $\text{St}(m, r) \times \text{St}(n, r)$. Under standard smoothness and lower-boundedness assumptions on f , the following theorem guarantees monotone descent and convergence to first-order stationarity on the product Stiefel manifold.

E.1 Proof of Theorem E.1

Proof. Let $\mathcal{M} = \text{St}(m, r) \times \text{St}(n, r)$ with the canonical product metric. For any $(U, V) \in \mathcal{M}$ define $W(U, V) := U\Sigma V^\top$. Since $\mathcal{M}$ is compact and $W(\cdot, \cdot)$ is smooth, the set

$$\mathcal{W} := \{W(U, V) : (U, V) \in \mathcal{M}\} \subset \mathbb{R}^{m \times n} \quad (76)$$

is compact. By assumption, ∇f is L_f -Lipschitz on $\mathbb{R}^{m \times n}$, hence on $\mathcal{W}$.

We claim that $\text{grad}F$ is Lipschitz on $\mathcal{M}$: there exists $L_F > 0$ such that for all $x, y \in \mathcal{M}$,

$$\|\text{grad}F(x) - \text{grad}F(y)\| \leq L_F \text{dist}(x, y), \quad (77)$$

where $\|\cdot\|$ is the norm induced by the metric and dist is the Riemannian distance on $\mathcal{M}$.

To see this, note that $F = f \circ W$ is smooth on the compact manifold $\mathcal{M}$, hence its Riemannian Hessian $\text{Hess}F(x) : T_x \mathcal{M} \rightarrow T_x \mathcal{M}$ exists and is continuous in x . Therefore the operator norm $x \mapsto \|\text{Hess}F(x)\|_{\text{op}}$ is continuous on $\mathcal{M}$ and attains a finite maximum:

$$L_F := \sup_{x \in \mathcal{M}} \|\text{Hess}F(x)\|_{\text{op}} < \infty. \quad (78)$$

For any unit-speed minimizing geodesic $\gamma : [0, 1] \rightarrow \mathcal{M}$ from x to y and any $t \in [0, 1]$,

$$\frac{d}{dt} \text{grad}F(\gamma(t)) = \text{Hess}F(\gamma(t))[\dot{\gamma}(t)]. \quad (79)$$

Taking norms and using $\|\dot{\gamma}(t)\| = \text{dist}(x, y)$ gives

$$\left\| \frac{d}{dt} \text{grad}F(\gamma(t)) \right\| \leq \|\text{Hess}F(\gamma(t))\|_{\text{op}} \|\dot{\gamma}(t)\| \leq L_F \text{dist}(x, y). \quad (80)$$

Integrating from $t = 0$ to 1 and using that parallel transport along γ preserves norms yields (77).

Let Retr be the polar retraction on each Stiefel factor, i.e., $\text{Retr}_U(\xi) = \text{msign}(U + \xi)$ and similarly for V , and let Retr denote the product retraction on $\mathcal{M}$. By smoothness of Retr , there exists a constant $c_R > 0$ such that for all $x \in \mathcal{M}$ and all $\xi \in T_x \mathcal{M}$ with $\|\xi\|$ small enough,

$$\text{dist}(\text{Retr}_x(\xi), \exp_x(\xi)) \leq c_R \|\xi\|^2, \quad (81)$$

where $\exp_x$ is the Riemannian exponential map. Fix $x \in \mathcal{M}$ and set $\xi = -\eta \text{grad}F(x)$. Define $y := \text{Retr}_x(\xi)$. Let γ be the minimizing geodesic from x to y . By the fundamental theorem of calculus on manifolds,

$$F(y) - F(x) = \int_0^1 \langle \text{grad}F(\gamma(t)), \dot{\gamma}(t) \rangle dt. \quad (82)$$

Add and subtract $\text{grad}F(x)$ inside the integrand:

$$F(y) - F(x) = \int_0^1 \langle \text{grad}F(x), \dot{\gamma}(t) \rangle dt + \int_0^1 \langle \text{grad}F(\gamma(t)) - \text{grad}F(x), \dot{\gamma}(t) \rangle dt. \quad (83)$$

The first term satisfies

$$\int_0^1 \langle \text{grad}F(x), \dot{\gamma}(t) \rangle dt = \langle \text{grad}F(x), \log_x(y) \rangle, \quad (84)$$

where $\log_x(y) \in T_x \mathcal{M}$ is the Riemannian logarithm and $\|\log_x(y)\| = \text{dist}(x, y)$. Using (81) and standard local properties of retractions, there exists $c_1 > 0$ such that

$$\|\log_x(y) - \xi\| \leq c_1 \|\xi\|^2. \quad (85)$$

Hence

$$\begin{aligned} \langle \text{grad}F(x), \log_x(y) \rangle &= \langle \text{grad}F(x), \xi \rangle + \langle \text{grad}F(x), \log_x(y) - \xi \rangle \\ &\leq -\eta \|\text{grad}F(x)\|^2 + c_1 \eta^2 \|\text{grad}F(x)\|^3. \end{aligned} \quad (86)$$

For the second term, by Cauchy–Schwarz, (77), and $\|\dot{\gamma}(t)\| = \text{dist}(x, y)$,

$$\begin{aligned} \int_0^1 \langle \text{grad}F(\gamma(t)) - \text{grad}F(x), \dot{\gamma}(t) \rangle dt &\leq \int_0^1 \|\text{grad}F(\gamma(t)) - \text{grad}F(x)\| \|\dot{\gamma}(t)\| dt \\ &\leq L_F \text{dist}(x, y)^2. \end{aligned} \quad (87)$$

Finally, since Retr is first-order, there exists $c_2 > 0$ such that $\text{dist}(x, y) \leq c_2 \|\xi\| = c_2 \eta \|\text{grad}F(x)\|$ for η small enough. Combining the bounds gives

$$F(y) \leq F(x) - \eta \|\text{grad}F(x)\|^2 + \left(c_1 \eta^2 \|\text{grad}F(x)\| + L_F c_2^2 \eta^2 \right) \|\text{grad}F(x)\|^2. \quad (88)$$

Because $\mathcal{M}$ is compact and $\text{grad}F$ is continuous, $\|\text{grad}F\|$ is bounded on $\mathcal{M}$, so the bracketed term is bounded above by $C\eta^2$ for some constant $C > 0$ independent of x . Thus there exists $\bar{L} > 0$ such that for all $\eta \in (0, 1/\bar{L}]$,

$$F(y) \leq F(x) - \frac{\eta}{2} \|\text{grad}F(x)\|^2. \quad (89)$$

Applying (89) with $x = (U_t, V_t)$ and $y = (U_{t+1}, V_{t+1})$ yields the stated descent inequality.

Summing (89) over $t = 0, \dots, T-1$ gives

$$\begin{aligned} \frac{\eta}{2} \sum_{t=0}^{T-1} \|\text{grad}F(U_t, V_t)\|^2 &\leq F(U_0, V_0) - F(U_T, V_T) \\ &\leq F(U_0, V_0) - \inf_{\mathcal{M}} F, \end{aligned} \quad (90)$$

and the right-hand side is finite since F is bounded below. Letting $T \rightarrow \infty$ yields $\sum_{t=0}^{\infty} \|\text{grad}F(U_t, V_t)\|^2 < \infty$, hence $\|\text{grad}F(U_t, V_t)\| \rightarrow 0$. Since $\mathcal{M}$ is compact, the sequence has accumulation points; continuity of $\text{grad}F$ implies that every accumulation point satisfies $\text{grad}F = 0$, that is, it is first-order stationary. $\square$

737 F Properties of SVD decompositions on weight matrices

738 **Lemma F.1** (SVD Decompositions with repeated singular values allow for rotations). *Let W denote*
 739 *a weight matrix, and $U\Sigma V^T$ be an SVD of W . Let the singular values of W be grouped into blocks*
 740 *of size $d_1, d_2, \dots, d_p$ of equal values. Then, for any block-diagonal orthogonal matrix*

$$R = \text{diag}(R_1, R_2, \dots, R_p), R_k \in O(d_k),$$

741 *the factorization $W = (UR)\Sigma(VR)^T$ is also a valid SVD of W .*

742 *Proof.* Let the columns of U be $u_1, \dots$ and the columns of V be $v_1, \dots$. We must first check that UR
 743 and VR have orthonormal columns, and then that the product $(UR)\Sigma(VR)^T$ equals W .

744 For the first assertion, we see that $U^T U = I$ because U comes from an SVD, and similarly $R^T R = I$.
 745 Hence, we compute

$$(UR)^T (UR) = R^T U^T U R = I.$$

746 Hence, it follows that UR has orthonormal columns. By an identical argument, VR also satisfies this
 747 property.

748 We will now prove that our product is equal to W . Let the distinct singular values of W be $\sigma_1 >$
 749 $\sigma_2 > \dots > \sigma_p$. We write $\Sigma = \text{diag}(\sigma_1 I_{d_1}, \sigma_2 I_{d_2}, \dots, \sigma_p I_{d_p})$.

750 Observe that

$$(UR)\Sigma(VR)^T = U(R\Sigma R^T)V.$$

751 Hence, we must show that $R\Sigma R^T = \Sigma$, after which our proof easily follows. Because both R and Σ
 752 are block-diagonal, we can compute our product block-wise. We have

$$\begin{aligned} R\Sigma R^T &= \text{diag}(R_1, \dots, R_p) \text{diag}(\sigma_1 I_{d_1}, \dots, \sigma_p I_{d_p}) \text{diag}(R_1^T, \dots, R_p^T) \\ &= \text{diag}(\sigma_1 R_1 I_{d_1} R_1^T, \dots, \sigma_p R_p I_{d_p} R_p^T) \\ &= \text{diag}(\sigma_1 I_{d_1}, \dots, \sigma_p I_{d_p}) \\ &= \Sigma. \end{aligned}$$

753 This completes our proof. □

754 **Lemma F.2** (SVD Decompositions can only differ by a rotation). *Let W denote a weight matrix, and*
 755 *$U\Sigma V^T$ be an SVD of W . Let the singular values of W be grouped into blocks of size $d_1, d_2, \dots, d_p$ of*
 756 *equal values. Then, if there exists another SVD $U'\Sigma V'^T = W$, we must have $U' = UR$ and $V' = VR$*
 757 *for some unique block-diagonal orthogonal matrix*

$$R = \text{diag}(R_1, R_2, \dots, R_p), R_k \in O(d_k).$$

758 *Proof.* We first claim that V' and V differ by an orthogonal matrix. We then claim that it must be
 759 block diagonal. Then, we will show that the same matrix relates U and U' . Finally, we will show that
 760 this matrix R is unique.

761 Let the not-necessarily-distinct singular values of W be $\tau_1 \geq \dots \geq \tau_r$. Let the distinct singular
 762 values of W be $\sigma_1 > \dots > \sigma_{d_p}$.

763 We first prove our first claim. We know that V and V' are both orthogonal. Observe that if $R = V^T V'$,
 764 then $R^T R = V'^T V V^T V' = I$. Hence, R is orthogonal. Furthermore, we have $V' = VR$, so we
 765 conclude that V and V' differ by an orthogonal matrix. This completes the proof of the first claim.

766 Next, observe that

$$W^T W = (U\Sigma V^T)^T (U\Sigma V^T) = (V\Sigma^T U^T)(U\Sigma V^T) = V\Sigma^2 V^T.$$

767 By a similar manipulation, we conclude that $W^T W = V \Sigma^2 V^T = V' \Sigma^2 V'^T$. Define e_i to be the i th
 768 standard basis vector. Then, it follows that

$$\begin{aligned} v_i &= V e_i \\ W^T W v_i &= V \Sigma^2 V^T (V e_i) \\ W^T W v_i &= V \Sigma^2 e_i \\ W^T W v_i &= V \sigma_i^2 = \tau_i^2 v_i. \end{aligned}$$

769 Hence, it follows that v_i is an eigenvector of $W^T W$ for all i . Therefore, we conclude that V is an
 770 orthonormal eigenbase of $W^T W$. By a similar argument, V' also satisfies this property.

771 Additionally, we have

$$\Sigma^2 = V^T W^T W V = R \Sigma^2 R^T.$$

772 Therefore, R lies in the commutant of Σ^2 .

773 We can write $\Sigma^2 = \text{diag}(\sigma_1^2 I_{d_1}, \dots, \sigma_p^2 I_{d_p})$ with $\sigma_1^2, \dots$ distinct. Then, R commutes if and only if it
 774 is block-diagonal with respect to this decomposition. This is true because if R had a non-zero entry
 775 mixing two blocks with different eigenvalues, then $\Sigma^2 R \neq R \Sigma^2$. Hence, $R = \text{diag}(R_{d_1}, \dots, R_{d_p})$
 776 for which $R_{d_i} \in O(d_i)$. We have completed our first two claims, i.e. we have shown that R is
 777 orthogonal, and block-diagonal.

778 We next show that $U' = UR$. We know that $WV = U\Sigma$ and $WV' = U'\Sigma$. Therefore,

$$\begin{aligned} (WV)R &= (U\Sigma)R \\ W(VR) &= U'\Sigma \\ U(\Sigma)R &= U'\Sigma. \end{aligned}$$

779 Because Σ is invertible on its support, we conclude that $UR = U'$.

780 To conclude, we show that R is unique. This follows because if $U' = UR = UR'$, then multiplying
 781 on the left by U^T yields $R = R'$. This completes the proof of our Lemma. $\square$

782 We conclude this appendix by proving mathematically that the existence of this rotation matrix R
 783 after reinforcement learning is not indicative of any form of learning.

784 **Theorem F.3** (Rotation matrices are not indicative of learning). *Let W denote the weight matrix of a
 785 model before reinforcement learning. Assume that the singular values of W contain clusters with
 786 small spectral gaps. Let $W' = W + \varepsilon E$ for some small ε where E is a random noise matrix. Define
 787 the SVDs $W = U\Sigma V$ and $W' = U'\Sigma'V'$. Then there exists a block-diagonal orthogonal matrix R
 788 such that $U' \approx UR$ and $V' \approx VR$, where the blocks of R correspond to clusters of singular values.
 789 Consequently, the existence of such a rotation R does not, by itself, constitute evidence of learning.*

790 *Proof.* Recall that $W^T W = V \Sigma^2 V^T$ and $W'^T W' = V' \Sigma'^2 V'^T$. Expanding, we have $W'^T W' =$
 791 $W^T W + \varepsilon(W^T E + E^T W) + O(\varepsilon^2)$.

792 By the Davis-Kahan theorem, we know that the eigenspace of $W^T W$ is stable. Hence, from our
 793 previous claims, there exists an orthogonal block-diagonal matrix R for which $V' \approx VR$. It follows
 794 from our previous claims that $U' \approx UR$.

795 Therefore, the existence of a matrix R that aligns (U, V) and (U', V') does not indicate that learning
 796 has taken place. Rather, it is just a consequence of linear algebra. $\square$

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964 Question: Does the paper report error bars suitably and correctly defined or other appropriate
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